Almost-perfect non-local quantum computation requires high-dimensional resources

Companion notes to arXiv:2609.40228

Marten Folkertsma, Dmitry Grinko, Gina Muuss, and Florian Speelman

QuSoft, University of Amsterdam

3 October 2026

Minor update: 22 October 2026

Status of these notes. This is an almost-entirely AI-written exposition of extensions of [FGMS26] to approximate protocols. We have spent time understanding the main proof, but have not simplified, reconstructed, and rewritten these arguments as we have for the main paper.
The main results also have AI-generated Lean proofs using only Lean's standard axioms. Differences from the written proofs are described in Section 9; the sources are at `https://github.com/fspeelman/NLQC/`.

Abstract

The main paper proves that almost every bipartite unitary, already on two qubits, has no exact non-local quantum computation protocol with finite-dimensional resources. These companion notes give lower bounds on the resources needed as the approximation error ε tends to zero.

We charge the footprint $K = r\, m_A m_B$, the resource Schmidt number times the dimensions of the two simultaneous messages. For n qubits per party, universal approximation of unitaries or rank-one projective measurements requires $\log_2 K \geq 2n + \frac{1}{2}\log_2 \ln(1/\varepsilon) - 6$ for $0 < \varepsilon \leq 1/2$. For almost every fixed target, below a target-dependent error threshold, the bound is $n + \frac{1}{2}\log_2 \ln(1/\varepsilon) - O(1)$. The errors are Choi entanglement infidelity for unitaries and average joint-label failure on basis inputs for measurements; the bounds also hold for normalized diamond and worst-case total-variation error, respectively. We obtain weaker precision bounds when classical messages are free, including for measurement localization, and extend the fixed-dimension lower bound to almost every diagonal gate.

The proof controls the overlap of compressed encoder and decoder isometries. Its derivative is tangent to the local-unitary orbit up to a leakage term, and a dimension-uniform tube estimate bounds the volume of approximately reachable targets. The main results, including the explicit universal constants and the effective bound for a named gate, are formalized in Lean 4 using only its standard axioms; Section 9 describes the coverage and proof differences.

1 Introduction

In non-local quantum computation (NLQC), two parties apply a joint operation to their quantum inputs using a shared entangled state and one simultaneous exchange of messages. The main paper [FGMS26] proves that Haar-almost every bipartite unitary, already on two qubits, has no exact protocol with finite-dimensional resources; the same holds for rank-one

projective measurements whose outcome both parties must learn. Every unitary nevertheless has arbitrarily accurate finite-resource approximations [Vai03, BCF+14, BK11]. These notes ask how fast the required resources grow as the error tends to zero.

We charge a protocol its *footprint* $K = r\, m_A m_B$, where r is the Schmidt number of its resource state and m_A, m_B are the message dimensions. Classical and quantum messages are charged alike; all private registers are free. Our bounds constrain Schmidt numbers and message dimensions, not entanglement entropy. The errors are the Choi entanglement infidelity δ_{Choi} for unitaries and the average probability δ_Φ that the parties do not both output the correct label on a uniformly sampled basis input for measurements. Normalized diamond distance and worst-case total-variation distance bound these errors from above (Section 2.2).

1.1 Main results

Let $d = 2^n$ be the local input dimension, with $n \geq 1$. The proofs also hold for every integer $d \geq 2$. Constants $c, C > 0$ and $O(1)$ are universal, independent of d, the footprint and the error. Write $K_\varepsilon(T)$ for the least footprint that approximates a target T with error at most ε. For any fixed target without an exact finite-resource protocol, compactness already gives $K_\varepsilon(T) \to \infty$ (Proposition 2.2); our first theorem gives rates for almost every target.

Theorem A (Haar volume; Theorem 5.5 and Corollaries 5.6 and 5.7)**.** *Let $n \geq 1$ and $0 < \varepsilon \leq 1/2$.*

(i) Volume. *For $K \geq d^2/4$, the unitaries that some protocol of footprint at most K implements with $\delta_{\mathrm{Choi}} \leq \varepsilon$ have Haar measure at most*

$$\exp(Cd^2K^2)\,\varepsilon^{\kappa_{\mathrm{u}}/2}, \qquad \kappa_{\mathrm{u}} = (d^2-1)(d^2-3).$$

The same holds for rank-one projective measurements with $\delta_\Phi \leq \varepsilon$, with the invariant measure on bases and exponent $\kappa_{\mathrm{m}} = (d^2-1)(d^2-2)$.

(ii) Universal approximation. *A footprint that suffices for every unitary, or for every rank-one projective measurement, satisfies*

$$K \geq c\, d\sqrt{\ln(1/\varepsilon)}, \qquad \textit{that is,} \qquad \log_2 K \geq n + \tfrac{1}{2}\log_2 \ln(1/\varepsilon) - O(1).$$

(iii) Almost every target. *For Haar-almost every $U \in \mathrm{U}(d^2)$ there is a threshold $\varepsilon_0(U) > 0$ such that $\log_2 K_\varepsilon(U) \geq n + \frac{1}{2}\log_2 \ln(1/\varepsilon) - O(1)$ for all $0 < \varepsilon < \varepsilon_0(U)$. The same holds for almost every rank-one projective measurement.*

The condition $K \geq d^2/4$ in part (i) holds whenever the footprint approximates SWAP or any basis measurement at error at most $1/2$ (Lemma B.1). The exponents are the codimension bounds from the exact argument. For universal approximation, we can improve the dimension factor by considering targets near SWAP, or near a maximally entangled basis.

Theorem B (Universal approximation; Corollaries 6.4 and 6.8)**.** *Let $n \geq 1$ and $0 < \varepsilon \leq 1/2$. A footprint K that suffices for every target satisfies*

$$\textit{Unitaries:} \qquad K \geq d^2 \max\left\{1-\varepsilon,\ \frac{\sqrt{\ln(1/\varepsilon)}}{51}\right\},$$

$$\textit{Rank-one PVMs:} \qquad K \geq \max\left\{d^3(1-\varepsilon)^2,\ \frac{d^2\sqrt{\ln(1/\varepsilon)}}{36}\right\}.$$

In both cases, $\log_2 K \geq 2n + \frac{1}{2}\log_2 \ln(1/\varepsilon) - 6$.

The distinction is between universal and generic bounds. Theorem A controls the whole target group using $O(d^2K^2)$ witness coordinates. The neighbourhoods used for Theorem B force both message dimensions to be of order d and permit smaller frozen environments, reducing this count to $O(K^2)$. Every universal budget must cover these neighbourhoods, but they do not give a stronger whole-group generic bound. The error-independent scale comes from elementary Schmidt-rank floors. Almost every fixed target also has the corresponding integer floor d^2 or d^3 below a target-dependent error threshold. The explicit constants make the precision cost quantitative; Appendix D records the calculation and improvements in larger dimensions.

The divergence with precision survives when classical communication is free.

Theorem C (Free classical communication; Lemma 7.2 and Corollaries 7.3 and 7.4)**.** *Suppose classical messages are free, possibly with continuous alphabets, and charge only the* quantum footprint $K_{\mathrm{q}} = r\, m_A^{\mathrm{q}} m_B^{\mathrm{q}}$, *where $m_A^{\mathrm{q}}, m_B^{\mathrm{q}}$ are the dimensions of the quantum parts of the messages. For $0 < \varepsilon \le 1/2$, a quantum footprint that suffices for every unitary, or for every rank-one projective measurement, satisfies*

$$\log_2 K_{\mathrm{q}} \ge \tfrac{1}{10} \log_2 \ln(1/\varepsilon) - \tfrac{2}{5} n - O(1),$$

and in addition $K_{\mathrm{q}} \ge d^2(1-\varepsilon)$ for unitaries and $K_{\mathrm{q}} \ge d(1-\varepsilon)$ for measurements. For almost every fixed unitary or measurement, below a target-dependent error threshold, the same precision bound holds with $\frac{3}{5}n$ in place of $\frac{2}{5}n$, and in addition $K_{\mathrm{q}} \ge d^2$ for unitaries and $K_{\mathrm{q}} \ge d$ for measurements. With only classical messages (local operations and simultaneous classical communication, LOSCC), $K_{\mathrm{q}} = r$ is the Schmidt number of the resource; in particular, localizing the measurement in a Haar-random basis without communication, to failure probability ε below a threshold, requires a resource of Schmidt number $r \ge \max\{d,\ c\, d^{-3/5}[\ln(1/\varepsilon)]^{1/10}\}$.

At fixed dimension, these results force double-logarithmic growth of $\log_2 K$, or of the number of initial resource qubits in LOSCC, for almost every target. Section 8 gives the same order within the Haar-null family of diagonal gates by following one scalar invariant; this is an application of the method rather than a consequence of the whole-group measure bound.

1.2 Proof idea and reading guide

The exact argument starts from a frozen dilation: if a protocol implements a unitary U exactly, its discarded environment is independent of the input. After compression, encoder and decoder isometries satisfy $\mathsf{Enc} = \mathsf{Dec}U$, and their infinitesimal generators are local. Differentiation confines the target to a local-unitary orbit; finitely many semialgebraic components then give finitely many orbits per architecture.

For an approximate protocol, we first freeze the environment up to error $O(\sqrt{\varepsilon})$, with Schmidt rank controlled by K (Section 3). We compress the uncharged registers and complete the reversed local decoders to isometries, obtaining a common finite-dimensional witness. Its overlap $\widehat{T} = \mathsf{Dec}^\dagger\mathsf{Enc}$ is close to the target and satisfies

$$\dot{\widehat{T}} = -b\widehat{T} + \widehat{T}a + \Xi, \qquad \|\Xi\|_F \le \sqrt{h}(\|\dot{\mathsf{Enc}}\|_{\mathrm{op}} + \|\dot{\mathsf{Dec}}\|_{\mathrm{op}}),$$

where $h = d^2 - \|\widehat{T}\|_F^2$ is the leakage (Section 4). The first two terms are local-unitary directions; for measurements the output generator is diagonal instead.

Section 5 bounds the transverse volume of these low-leakage witness images by a direct polynomial image-volume argument, using Lagrange multipliers, Sard's theorem and an elementary count of isolated polynomial zeros. Comparing $O(d^2K^2)$ parameters with $\Theta(d^4)$ transverse target directions proves Theorem A. Section 6 gives the smaller witness families for Theorem B. Section 7 then compresses even continuous classical transcripts to finite ones without decreasing

the score, and applies the charged-message bounds to prove Theorem C. Readers interested in the diagonal-gate application can proceed from the witness construction to Section 8, which uses the scalar image-volume lemma rather than the tube estimate. Appendices A and B contain the geometric details and elementary rank floors; Appendix D records the explicit universal constants.

Section 9 summarizes the Lean coverage and the differences from the written proofs; the declaration table is in the repository documentation. Appendix C gives an effective bound for the named gate $C_1 = \mathrm{diag}(1, 1, 1, e^i)$, namely $\log_2 K_\varepsilon(C_1) \geq \frac{1}{2} \log_2 \ln \ln(1/\varepsilon) - O(1)$. That arithmetic argument is independent of the geometric precision bounds developed here.

1.3 Related work

Upper bounds. Every bipartite unitary can be approximated in NLQC [Vai03, BCF+14]; the port-based construction of Beigi and König [BK11] with the bounds of [CLM+21] uses $O(n\, 2^{4n}/\varepsilon)$ ebits for diamond error ε, and $O(\varepsilon^{-1/2})$ ebits at fixed n. For controlled unitaries and two-qubit unitaries, $O(\log(1/\varepsilon))$ ebits suffice [Yu11, YGC12, GC20]. For comparison, at fixed n our lower bounds on $\log_2 K$, and on the number of resource qubits when only classical messages are sent, are of order $\log \log(1/\varepsilon)$.

Lower bounds. Most lower bounds on NLQC concern the growth of the resources with the input size at a fixed error, often motivated by quantum position verification [BK11, TFKW13, BCS22, ACCM25]. Rank and entanglement lower bounds are known for f-routing and unitary implementation [ACM25, CM26, Bog26]; for instance, Bogner proves an $\Omega(\log n)$ entanglement-of-formation lower bound for inner-product f-routing at constant error [Bog26]. Using the geometry of Banach spaces, exponential resource lower bounds have been proved for restricted classes of strategies [JKPPG22], recently for strategies whose first round has low degree as a function of the classical inputs [MCPPG26]. The latter hypothesis concerns the whole family of first-round operations across inputs, not a single target; it gives family-wise hardness rather than a fixed target that is hard for every protocol. Our bounds are complementary: they concern the dependence on the precision at a given input size, uniformly in n, and for a single target they apply to every protocol. At constant error our methods give only the elementary floors, of order d^2 for universal approximation of unitaries and d^3 for measurements.

Methods. Approximate compression of infinitesimal generators underlies the approximate Eastin–Knill theorems [FNA+20]. Dimension counting for the semialgebraic set of unitaries reachable by a fixed circuit architecture gives linear growth of exact circuit complexity [HFK+22]; robust versions use different tools [Haf23]. Our tube estimate is a dimension-uniform variant of quantitative Sard estimates in tame geometry [YC04]. We prove the needed polynomial image-volume bound directly by selecting fibrewise critical points and counting isolated zeros of their polynomial equations.

2 Setting

We use the protocol model and the notation of [FGMS26], which we briefly recall, and then introduce the error measures and the resource measure used in these notes.

2.1 Protocols and exact rigidity

Alice and Bob receive the input registers A and B, with $H_A \cong H_B \cong \mathbb{C}^d$ and $d = 2^n$. They share a resource state on $R_A R_B$, apply local encoding isometries

$$V_A : AR_A \to K_A M_{A_{\mathrm{pre}}}, \qquad V_B : BR_B \to K_B M_{B_{\mathrm{pre}}},$$

exchange the message registers with the exchange unitary X_{ex}, which sends $M_{A_{\mathrm{pre}}}$ to $M_{B_{\mathrm{post}}}$ and $M_{B_{\mathrm{pre}}}$ to $M_{A_{\mathrm{post}}}$, and apply local decoding isometries

$$W_A : K_A M_{A_{\mathrm{post}}} \to A' G_A, \qquad W_B : K_B M_{B_{\mathrm{post}}} \to B' G_B.$$

Auxiliary registers of the Stinespring dilations are included in the kept registers K_A, K_B and the garbage registers G_A, G_B, as in [FGMS26]. The message dimensions are

$$m_A = \dim M_{A_{\mathrm{pre}}} = \dim M_{B_{\mathrm{post}}}, \qquad m_B = \dim M_{B_{\mathrm{pre}}} = \dim M_{A_{\mathrm{post}}}.$$

For a pure resource $|\eta\rangle$, the protocol is the isometry

$$F = (W_A \otimes W_B)\, X_{\mathrm{ex}}\, (V_A \otimes V_B)\, E_\eta : AB \to A' G_A B' G_B, \tag{1}$$

where E_v is the insertion map $E_v \psi = \psi \otimes v$, and it implements the channel $\mathcal{N}_F(\rho) = \mathrm{Tr}_{G_A G_B}(F \rho F^\dagger)$.

Mixed resources. In [FGMS26] the resource is a pure state. For exact implementation this is no restriction, since a purifying register can be given to one party, but purification can increase the Schmidt rank, and the Schmidt rank is one of the resources we charge. We therefore allow a mixed resource σ on $R_A R_B$. Its *Schmidt number* is the least r such that σ is a convex combination of pure states of Schmidt rank at most r. For such a decomposition $\sigma = \sum_j p_j |\eta_j\rangle\langle\eta_j|$, the protocol implements the average $\sum_j p_j \mathcal{N}_{F_j}$ of the channels of the pure protocols F_j with resource $|\eta_j\rangle$. The Choi infidelity and the failure probability of Section 2.2, which our lower bounds use, are affine in the implemented channel, so some pure component has at least the average score; this is how we reduce to pure resources. Shared classical randomness is part of the resource: a seed j known to both parties, together with an entangled state $|\eta_j\rangle$, is the classically flagged mixture $\sum_j p_j |jj\rangle\langle jj| \otimes |\eta_j\rangle\langle\eta_j|$, whose Schmidt number is at most $\max_j \mathrm{SR}(\eta_j)$, and selecting a pure component fixes the seed. Purifying the seed coherently would instead increase the Schmidt rank. Local instruments with classical outcomes and local randomness are included by coherent dilation: outcomes are recorded coherently, later operations are controlled on them, and copies of dephased labels are placed in the garbage registers.

Measurements. For a rank-one projective measurement (PVM) $\Phi = (P_1, \ldots, P_{d^2})$ with $P_i = |\varphi_i\rangle\langle\varphi_i|$, the output registers A', B' are label registers of dimension d^2, read out in the basis $|1\rangle, \ldots, |d^2\rangle$, and both parties must output the correct label. The *basis matrix* $M = \sum_i |\varphi_i\rangle\langle i|$ is determined by Φ up to $M \mapsto M\Delta$ with Δ diagonal unitary.

Local orbits. Write $\mathcal{L} = \{L_A \otimes L_B : L_A, L_B \in \mathrm{U}(d)\}$ for the local unitary group. As in [FGMS26], the orbit of a unitary and the orbit of a basis matrix are

$$\mathfrak{o}(U) = \{LUR : L, R \in \mathcal{L}\}, \qquad \mathfrak{o}_{\mathrm{m}}(M) = \{LM\Delta : L \in \mathcal{L},\ \Delta \in \mathrm{U}(d^2) \text{ diagonal}\},$$

and the corresponding infinitesimal generators are the local anti-Hermitian operators

$$\mathfrak{u}_{\mathrm{loc}} = \{a_A \otimes I + I \otimes a_B : a_A^\dagger = -a_A,\ a_B^\dagger = -a_B\}$$

and the diagonal anti-Hermitian matrices $\mathfrak{u}_{\mathrm{diag}}$.

Exact rigidity. The exact proof starts from a frozen dilation. If F implements U exactly, then $F = E_\gamma U$ for a unit vector $|\gamma\rangle \in G_A G_B$, by the rigidity lemma of [FGMS26]. If F implements Φ exactly, then $F = C_\gamma M^\dagger$, where $C_\gamma|i\rangle = |i\rangle_{A'}|i\rangle_{B'}|\gamma_i\rangle$ is the duplicated-label map of unit vectors $|\gamma_1\rangle, \dots, |\gamma_{d^2}\rangle \in G_A G_B$, by its measurement version [FGMS26]. To treat both cases at once we write

$$(\Gamma, T) = (E_\gamma, U) \text{ for unitaries}, \qquad (\Gamma, T) = (C_\gamma, M^\dagger) \text{ for measurements}, \tag{2}$$

so that exactness reads $F = \Gamma T$ with $T \in \mathrm{U}(d^2)$. With

$$\mathsf{Enc} = X_{\mathrm{ex}}(V_A \otimes V_B)E_\eta, \qquad \mathsf{Dec} = (W_A \otimes W_B)^\dagger \Gamma,$$

exactness gives $\mathsf{Enc} = \mathsf{Dec}T$, and along a differentiable path of exact strategies [FGMS26]

$$\dot{T} = -b\,T + T\,a, \qquad a = \mathsf{Enc}^\dagger \dot{\mathsf{Enc}} \in \mathfrak{u}_{\mathrm{loc}}, \qquad b = \mathsf{Dec}^\dagger \dot{\mathsf{Dec}} \in \begin{cases} \mathfrak{u}_{\mathrm{loc}} & \text{for unitaries,} \\ \mathfrak{u}_{\mathrm{diag}} & \text{for measurements.} \end{cases} \tag{3}$$

For measurements this is the adjoint of the identity $\dot{M} = -aM + Mb$ of [FGMS26].

For an arbitrary complex $d^2 \times d^2$ matrix Y we call

$$\mathcal{T}_{\mathrm{u}}(Y) = \{-bY + Ya : a, b \in \mathfrak{u}_{\mathrm{loc}}\}, \qquad \mathcal{T}_{\mathrm{m}}(Y) = \{-bY + Ya : a \in \mathfrak{u}_{\mathrm{loc}},\ b \in \mathfrak{u}_{\mathrm{diag}}\} \tag{4}$$

the *allowed directions* at Y. At a unitary $Y = U$, $\mathcal{T}_{\mathrm{u}}(U)$ is the tangent space of $\mathfrak{o}(U)$; at $Y = M^\dagger$, $\mathcal{T}_{\mathrm{m}}(M^\dagger)$ is the tangent space of $\{M'^\dagger : M' \in \mathfrak{o}_{\mathrm{m}}(M)\}$. The real dimension of $\mathfrak{u}_{\mathrm{loc}}$ is $2d^2 - 1$: the map $(a_A, a_B) \mapsto a_A \otimes I + I \otimes a_B$ on pairs of anti-Hermitian matrices, a space of real dimension $2d^2$, has the one-dimensional kernel spanned by $(iI, -iI)$. At every Y, the kernel of the map $(a, b) \mapsto -bY + Ya$, on $\mathfrak{u}_{\mathrm{loc}} \oplus \mathfrak{u}_{\mathrm{loc}}$ and on $\mathfrak{u}_{\mathrm{loc}} \oplus \mathfrak{u}_{\mathrm{diag}}$ alike, contains the line spanned by (iI, iI). Hence, at every matrix Y,

$$\dim_{\mathbb{R}} \mathcal{T}_{\mathrm{u}}(Y) \le \ell_{\mathrm{u}} := 4d^2 - 3, \qquad \dim_{\mathbb{R}} \mathcal{T}_{\mathrm{m}}(Y) \le \ell_{\mathrm{m}} := 3d^2 - 2, \tag{5}$$

in agreement with the orbit dimensions counted in [FGMS26]. The group $\mathrm{U}(d^2)$ has real dimension $N := d^4$, and we write

$$\kappa_{\mathrm{u}} := N - \ell_{\mathrm{u}} = (d^2 - 1)(d^2 - 3), \qquad \kappa_{\mathrm{m}} := N - \ell_{\mathrm{m}} = (d^2 - 1)(d^2 - 2) \tag{6}$$

for the corresponding codimensions; since ℓ bounds the dimension of every orbit, κ is a lower bound on the codimension of every orbit. Both are at least $\frac{3}{16}d^4$ for every $d \ge 2$. When a statement applies to both target classes, we drop the subscripts and write $\mathcal{T}$, ℓ and κ.

2.2 Error measures

Let $\bar{A} \cong A$ and $\bar{B} \cong B$ be reference registers and

$$|\Omega\rangle = \frac{1}{d}\sum_{j=1}^{d^2} |j\rangle_{AB}|j\rangle_{\bar{A}\bar{B}}$$

the maximally entangled state. For an operator X with domain AB we write $|X\rangle\!\rangle = (X \otimes I_{\bar{A}\bar{B}})|\Omega\rangle$ for its normalized Choi vector. Then

$$\|X\|_{\mathrm{n}} := \frac{1}{d}\|X\|_F = \||X\rangle\!\rangle\| \tag{7}$$

is the *normalized Frobenius norm*, which gives unitaries norm one. For two isometries F, F' with the same domain and codomain, $\|F - F'\|_{\mathrm{n}}^2 = 2 - 2d^{-2}\,\mathrm{Re}\,\mathrm{Tr}(F'^\dagger F)$.

Unitaries. The Choi state of a channel $\mathcal{N}$ is $\rho_{\mathcal{N}} = (\mathcal{N} \otimes \mathrm{id})(|\Omega\rangle\langle\Omega|)$. The *score* of $\mathcal{N}$ for a target unitary U and its *Choi infidelity* are

$$q_U(\mathcal{N}) = \langle\langle U|\rho_{\mathcal{N}}|U\rangle\rangle, \qquad \delta_{\mathrm{Choi}}(\mathcal{N}, U) = 1 - q_U(\mathcal{N}). \tag{8}$$

Thus q_U is the fidelity of the Choi state with the pure target Choi state, and δ_{Choi} is the entanglement infidelity. For the channel $\mathcal{N}_F$ of a pure protocol, $q_U(\mathcal{N}_F) = \|(\langle\langle U| \otimes I_{G_A G_B})|F\rangle\rangle\|^2$. The normalized diamond distance $\delta_\diamond(\mathcal{N}, U) = \frac{1}{2}\|\mathcal{N} - \mathcal{U}\|_\diamond$, with $\mathcal{U}(\rho) = U\rho U^\dagger$, satisfies

$$\delta_{\mathrm{Choi}}(\mathcal{N}, U) \le \delta_\diamond(\mathcal{N}, U), \tag{9}$$

because $1 - \langle\langle U|\rho_{\mathcal{N}}|U\rangle\rangle$ is at most the trace distance between $\rho_{\mathcal{N}}$ and the pure state $\rho_{\mathcal{U}} = |U\rangle\rangle\langle\langle U|$. All our lower bounds only use $\delta_{\mathrm{Choi}} \le \varepsilon$, and therefore also hold under $\delta_\diamond \le \varepsilon$.

Measurements. Let $p_\rho(a, b)$ be the probability that Alice outputs a and Bob outputs b on input ρ. The score of a protocol for the PVM Φ and its *failure probability* are

$$q_\Phi = \frac{1}{d^2}\sum_{i=1}^{d^2} p_{\varphi_i}(i, i), \qquad \delta_\Phi = 1 - q_\Phi. \tag{10}$$

This is the probability that both parties output the correct label when the input is a uniformly random basis vector; it does not depend on the phases of the φ_i. The worst-case total-variation error $\sup_\rho \frac{1}{2}\sum_{a,b}|p_\rho(a, b) - \delta_{ab}\,\mathrm{Tr}(P_a\rho)|$ is at least δ_Φ, as one sees by taking $\rho = |\varphi_i\rangle\langle\varphi_i|$ and averaging over i.

2.3 The footprint

Definition 2.1 (Footprint)**.** *A protocol has* footprint at most K *if its resource has Schmidt number r and its messages have dimensions m_A, m_B with*

$$r\, m_A m_B \le K.$$

The footprint charges the full dimension of each message, whether the message is quantum or classical; a classical message with m possible values is a message register of dimension m. The kept registers K_A, K_B (not to be confused with the footprint K), the garbage registers G_A, G_B and all auxiliary registers are free; the proofs compress them. For exact protocols with finite-dimensional resources, classical and quantum messages are equivalent, since quantum messages can be teleported with additional finite entanglement [FGMS26]. This conversion changes the footprint, so quantitatively the two models differ; Section 7 treats the model in which classical messages are free.

For $K \ge 1$ and $\varepsilon \ge 0$, let $\mathsf{Reach}^{\mathrm{u}}_{K,\varepsilon} \subseteq \mathrm{U}(d^2)$ be the set of unitaries U for which some protocol of footprint at most K has $\delta_{\mathrm{Choi}} \le \varepsilon$, and let $\mathsf{Reach}^{\mathrm{m}}_{K,\varepsilon} \subseteq \mathrm{U}(d^2)$ be the set of matrices $M^\dagger$, where M is a basis matrix of a PVM that some protocol of footprint at most K implements with $\delta_\Phi \le \varepsilon$. The second set is invariant under $M^\dagger \mapsto \Delta M^\dagger$ for diagonal unitaries Δ, so its Haar measure equals the invariant measure of the corresponding set of PVMs in $\mathrm{U}(d^2)/\mathrm{U}(1)^{d^2}$. When a statement applies to both target classes we drop the superscripts and write $\mathsf{Reach}_{K,\varepsilon}$. For a target T we write

$$K_\varepsilon(T) = \min\{K : T \in \mathsf{Reach}_{K,\varepsilon}\}$$

for its least footprint at error ε, with the appropriate reachable set.

2.4 Qualitative divergence

The exact results of [FGMS26], together with compactness, already imply that the footprint must diverge for targets without an exact protocol.

Proposition 2.2 (Qualitative divergence)**.** *Fix $K \geq 1$ and a target, either a unitary U or a PVM Φ. Among protocols of footprint at most K, the least error $\varepsilon_K(T)$ is attained, and $\varepsilon_K(T) = 0$ if and only if the target has an exact protocol of footprint at most K. The sets $\mathsf{Reach}_{K,\varepsilon}$ are compact. Consequently, if the target has no exact protocol with finite-dimensional resources, then $K_\varepsilon(T) \to \infty$ as $\varepsilon \to 0$.*

Proof. The error is affine in the resource state, so we may restrict to pure resources of Schmidt rank at most r, and hence to $R_A \cong R_B \cong \mathbb{C}^r$. By Lemma 4.1 below, the kept registers can be taken of dimension $k_X \leq drm_X$, so the decoders act on spaces of dimension at most dK. The error only depends on the channel $\mathrm{Tr}_{G_X}(W_X \cdot W_X^\dagger)$ implemented by each decoder, where $X \in \{A, B\}$, and a channel from a space of dimension q to one of dimension q' has a Stinespring isometry with environment dimension at most qq'. Thus finitely many choices of register dimensions suffice, and for each of them the protocols form a compact set, a product of unit spheres and sets of isometries. The error is a continuous function of the target and the protocol, so its minimum is attained, and the set of targets with error at most ε is a finite union of projections of compact sets, hence compact.

If a unitary protocol has error zero, its Choi state has fidelity one with the pure state $|U\rangle\!\rangle\langle\!\langle U|$, so the two are equal and the channel is $\rho \mapsto U\rho U^\dagger$. If a measurement protocol has error zero, let E_{ab} be the effect on the input that leads to the outputs (a, b). Then $\langle\varphi_i|E_{ii}|\varphi_i\rangle = 1$ for all i, so positivity and $\sum_{a,b} E_{ab} = I$ give $E_{ab}|\varphi_i\rangle = 0$ for $(a, b) \neq (i, i)$; hence $E_{ab} = 0$ for $a \neq b$ and $E_{ii} = |\varphi_i\rangle\langle\varphi_i|$, which is exactness. Finally, if no exact protocol exists, then $\varepsilon_K(T) > 0$ for every K, and $K_\varepsilon(T) > K$ whenever $\varepsilon < \varepsilon_K(T)$. □

At fixed budget, the semialgebraic structure also gives a power-law bound on the distance to the exact target set.

Corollary 2.3 (Fixed-budget stability)**.** *Fix d, K and suppose $E_{d,K} := \mathsf{Reach}_{K,0}$ is nonempty. There are $C_{d,K}, \alpha_{d,K} > 0$ such that, for every $\varepsilon \geq 0$ and $T \in \mathsf{Reach}_{K,\varepsilon}$,*

$$\inf_{S \in E_{d,K}} \|T - S\|_{\mathrm{n}} \leq C_{d,K} \varepsilon^{\alpha_{d,K}}.$$

The set $E_{d,K}$ is a finite union of exact local-equivalence orbits.

Proof. The compact protocol spaces in the proof of Proposition 2.2 are semialgebraic, and their scores are polynomial in real and imaginary coordinates. Compact minimization and real quantifier elimination therefore make $g(T) := \varepsilon_K(T)$ continuous and semialgebraic, with zero set $E_{d,K}$. The distance $h(T) := \inf_{S \in E_{d,K}} \|T - S\|_{\mathrm{n}}$ is likewise continuous and semialgebraic. The semialgebraic Łojasiewicz inequality [BCR98, Corollary 2.6.7] on $\mathrm{U}(d^2)$ gives $h \leq C_{d,K} g^{\alpha_{d,K}}$, proving the bound since $g(T) \leq \varepsilon$. The orbit assertion follows from the exact finite-orbit theorem [FGMS26] and the finite collection of register dimensions above. □

Neither result controls the dependence on K; in particular, the stability constants and exponent may depend on the budget. The remaining sections supply explicit resource bounds.

3 Approximately frozen dilations

The exact proof starts from the observation that the environment of an exact protocol ends in a fixed state. For an accurate protocol, we now construct a nearby frozen dilation whose environment state has Schmidt rank bounded by the footprint. This bound is what later makes the number of parameters independent of the uncharged garbage registers.

3.1 Schmidt rank across the laboratory cut

The Choi vector $|F\rangle\!\rangle$ of a protocol lies in $A'G_AB'G_B\bar{A}\bar{B}$. The *laboratory cut* separates Alice's systems $A'G_A\bar{A}$ from Bob's systems $B'G_B\bar{B}$; for vectors in G_AG_B we use the cut between G_A and G_B. For a vector χ in a bipartite space, not necessarily normalized, let $\mathrm{SR}(\chi)$ be its Schmidt rank, let $\lambda_1 \geq \lambda_2 \geq \dots$ be its squared Schmidt coefficients (padded with zeros), and let

$$\Lambda_k(\chi) = \sum_{j=1}^{k} \lambda_j = \max\{|\langle\chi' \mid \chi\rangle|^2 : \|\chi'\| = 1,\ \mathrm{SR}(\chi') \leq k\}, \qquad \Lambda_0(\chi) = 0, \tag{11}$$

be its *top-k Schmidt mass*; thus $\Lambda_k(\chi) \leq \|\chi\|^2$, with equality for $k \geq \mathrm{SR}(\chi)$. The variational formula is the Eckart–Young theorem for the best rank-k approximation of the coefficient matrix of χ in Frobenius norm [Bha97]. For unit vectors χ and χ', each in a bipartite space, the product $\chi \otimes \chi'$ has the squared Schmidt coefficients $\lambda_i(\chi)\lambda_j(\chi')$, and

$$\Lambda_k(\chi \otimes \chi') \leq \min\{\Lambda_k(\chi), \Lambda_k(\chi')\}. \tag{12}$$

Indeed, any k pairs (i, j) involve at most k distinct indices j, and grouping the pairs by j bounds their mass by $\Lambda_k(\chi')$; the other bound is symmetric.

Lemma 3.1 (Schmidt rank of a protocol)**.** *If the resource $|\eta\rangle$ has Schmidt rank r, then $\mathrm{SR}(|F\rangle\!\rangle) \leq r\, m_A m_B$ across the laboratory cut.*

Proof. Before the protocol, the input registers are maximally entangled with the references, A with $\bar{A}$ and B with $\bar{B}$, and the state of $A\bar{A}R_A$ versus $B\bar{B}R_B$ has Schmidt rank r. The encoders are local isometries and do not change the Schmidt rank. Moving a register of dimension m from one laboratory to the other multiplies the Schmidt rank by at most m: expanding that register in a basis writes every Schmidt term as a sum of m product terms across the new cut. The exchange moves $M_{A_{\mathrm{pre}}}$ and $M_{B_{\mathrm{pre}}}$, and the decoders are again local. □

3.2 Approximately frozen dilations

Lemma 3.2 (Approximately frozen dilations)**.** *Let a protocol have footprint at most K, and let $0 \leq \varepsilon < 1$.*

(i) If it implements U with $\delta_{\mathrm{Choi}} \leq \varepsilon$, then some pure component of its resource gives a protocol isometry F and a unit vector $|\gamma\rangle \in G_AG_B$ with

$$\mathrm{SR}(\gamma) \leq K, \qquad \|F - E_\gamma U\|_{\mathrm{n}} \leq \sqrt{2\varepsilon}. \tag{13}$$

(ii) If it implements Φ with $\delta_\Phi \leq \varepsilon$, then some pure component gives a protocol isometry F and unit vectors $|\gamma_1\rangle, \dots, |\gamma_{d^2}\rangle \in G_AG_B$ with

$$\sum_{i=1}^{d^2} \mathrm{SR}(\gamma_i) \leq K + d^2, \qquad \|F - C_\gamma M^\dagger\|_{\mathrm{n}} \leq 2\sqrt{\varepsilon}. \tag{14}$$

In the notation (2), both parts say that $\|F - \Gamma T\|_{\mathrm{n}} \leq \delta$ with

$$\delta = \sqrt{2\varepsilon} \text{ for unitaries}, \qquad \delta = 2\sqrt{\varepsilon} \text{ for measurements}. \tag{15}$$

At $\varepsilon = 0$ the lemma reduces to the rigidity lemma of [FGMS26] and its measurement version, with the additional information that the environment can be taken of Schmidt rank at most K (respectively $K + d^2$).

Proof. Both scores are affine in the resource state, so a decomposition witnessing Schmidt number at most r contains a pure component with score at least $1-\varepsilon$. Fix it and let $v = |F\rangle\!\rangle$. By Lemma 3.1, v is a unit vector with $\mathrm{SR}(v) \le K$.

(i) Project onto the target Choi vector: let $\xi = (\langle\!\langle U| \otimes I_{G_A G_B})v \in G_A G_B$. The score is $p := \|\xi\|^2 = q_U \ge 1-\varepsilon > 0$, and we put $\gamma' = \xi/\sqrt{p}$. The unit vector $|U\rangle\!\rangle \otimes \gamma'$ has overlap $\sqrt{p}$ with v, so (11) with $\mathrm{SR}(v) \le K$ and (12) give

$$p \le \Lambda_K(|U\rangle\!\rangle \otimes \gamma') \le \Lambda_K(\gamma') =: \tau.$$

Let γ be the normalized truncation of γ' to its K largest Schmidt terms, so that $\mathrm{SR}(\gamma) \le K$ and $\langle \gamma \mid \gamma' \rangle = \sqrt{\tau}$. The Choi vector of $E_\gamma U$ is $|U\rangle\!\rangle \otimes \gamma$, and

$$(\langle\!\langle U| \otimes \langle\gamma|)v = \langle \gamma \mid \xi \rangle = \sqrt{p\tau} \ge p.$$

Both v and $|U\rangle\!\rangle \otimes \gamma$ are unit vectors, so $\|F - E_\gamma U\|_{\mathrm{n}}^2 \le 2 - 2p \le 2\varepsilon$.

(ii) Write $|\Omega\rangle = d^{-1}\sum_i |\varphi_i\rangle|\bar{\varphi}_i\rangle$, with $\bar{\varphi}_i$ the complex conjugate of φ_i in the reference. Let Π be the projector onto the mutually orthogonal subspaces $|i\rangle_{A'}|i\rangle_{B'}|\bar{\varphi}_i\rangle_{\bar{A}\bar{B}} \otimes G_A G_B$. Then

$$v_0 := \Pi v = \frac{1}{d}\sum_{i=1}^{d^2} |i\rangle_{A'}|i\rangle_{B'}|\bar{\varphi}_i\rangle|\tilde{\gamma}_i\rangle, \qquad \tilde{\gamma}_i = (\langle i|_{A'}\langle i|_{B'} \otimes I)F|\varphi_i\rangle, \tag{16}$$

with $\|\tilde{\gamma}_i\| \le 1$, and the score is $p := \|v_0\|^2 = d^{-2}\sum_i \|\tilde{\gamma}_i\|^2 = q_\Phi \ge 1-\varepsilon$. Since $\langle v \mid v_0 \rangle = p$ and v is a unit vector of Schmidt rank at most K, (11) applied to the unnormalized vector v_0 gives $\Lambda_K(v_0) \ge p^2$.

The two copies of the label make the Schmidt decomposition of v_0 block diagonal: the terms with different i are orthogonal on both sides of the laboratory cut. If ζ_{ik}^2 are the squared Schmidt coefficients of $\tilde{\gamma}_i$ and λ_{ij} those of the unit vector φ_i, the squared Schmidt coefficients of v_0 are

$$d^{-2}\,\zeta_{ik}^2\,\lambda_{ij}. \tag{17}$$

Select K triples (i,j,k) that carry the top-K mass of v_0, and let $\hat{\gamma}_i$ be the part of the Schmidt decomposition of $\tilde{\gamma}_i$ consisting of the terms k that occur in a selected triple. Since $\sum_j \lambda_{ij} = 1$, these vectors satisfy

$$\sum_i \mathrm{SR}(\hat{\gamma}_i) \le K, \qquad d^{-2}\sum_i \|\hat{\gamma}_i\|^2 \ge \Lambda_K(v_0) \ge p^2.$$

Put $t_i = \|\hat{\gamma}_i\| \le 1$. If $t_i > 0$, let $\gamma_i = \hat{\gamma}_i/t_i$; orthogonality of the Schmidt terms gives $\langle \gamma_i \mid \tilde{\gamma}_i \rangle = t_i$. If $t_i = 0$, let γ_i be a unit product vector, chosen as a Schmidt term of $\tilde{\gamma}_i$ with a phase that makes $\langle \gamma_i \mid \tilde{\gamma}_i \rangle \ge 0$ if $\tilde{\gamma}_i \ne 0$. This adds at most one Schmidt term per outcome, so $\sum_i \mathrm{SR}(\gamma_i) \le K + d^2$. The Choi vector of $C_\gamma M^\dagger$ is $d^{-1}\sum_i |i\rangle|i\rangle|\bar{\varphi}_i\rangle|\gamma_i\rangle$, a unit vector, and

$$\mathrm{Re}\langle |C_\gamma M^\dagger\rangle\!\rangle, v\rangle = \frac{1}{d^2}\sum_i \mathrm{Re}\langle \gamma_i \mid \tilde{\gamma}_i\rangle \ge \frac{1}{d^2}\sum_i t_i \ge \frac{1}{d^2}\sum_i t_i^2 \ge p^2.$$

Hence $\|F - C_\gamma M^\dagger\|_{\mathrm{n}}^2 \le 2 - 2p^2 \le 4\varepsilon$. □

Remark 3.3 (Which error the proof uses). The proof only uses the score of the selected pure component. That component need not satisfy the operational error bound of the mixed protocol, for instance in diamond distance, but its score suffices for every later step. This is why all our lower bounds are stated for the scores, and why they transfer to the free classical model of Section 7, where the compression does not decrease the score.

4 The perturbed derivative identity

We now construct, for each accurate protocol, two isometries with a common codomain whose overlap is close to the target, and prove that the derivative of this overlap consists of allowed directions plus a small leakage term. The construction uses only coordinate spaces whose dimensions are bounded in terms of K and d.

4.1 A common finite-dimensional witness

First we remove the dimensions of the kept registers that a protocol never uses. For a pure resource of Schmidt rank r we restrict R_A and R_B to the supports of the reduced states, of dimension r.

Lemma 4.1 (Reachable-space compression)**.** *The kept registers can be replaced by subspaces $\widetilde{K}_A \subseteq K_A$ and $\widetilde{K}_B \subseteq K_B$ with $\dim \widetilde{K}_X \le d\,r\,m_X$ for $X \in \{A, B\}$, without changing F. The decoder inputs $Q_A := \widetilde{K}_A M_{A_{\mathrm{post}}}$ and $Q_B := \widetilde{K}_B M_{B_{\mathrm{post}}}$ then have dimension at most $d\,r\,m_A m_B \le dK$.*

Proof. The domain of V_A has dimension at most dr. Let $\widetilde{K}_A$ be the span of the vectors $(I_{K_A} \otimes \langle\mu|)V_A|\psi\rangle$, where $|\psi\rangle$ runs over a basis of the domain and $|\mu\rangle$ over a basis of $M_{A_{\mathrm{pre}}}$; it has dimension at most drm_A, and the range of V_A lies in $\widetilde{K}_A M_{A_{\mathrm{pre}}}$. After the exchange, Alice's decoder only acts on $\widetilde{K}_A M_{A_{\mathrm{post}}}$, of dimension at most $drm_A m_B$. The same holds for Bob. □

This is the compression argument of [FGMS26], which there shows that only the Schmidt rank and the message dimensions matter for the finite-orbit theorem. We then pad the dimensions to fixed values: the resource registers to $\mathbb{C}^r$ and the kept registers to dimension drm_X, so that the decoder inputs Q_X have dimension $drm_A m_B \le dK$. An encoder extends isometrically to added resource directions by completing its orthonormal columns, which is possible because its padded codomain has dimension $drm_X^2 \ge dr$, and a decoder extends to added input directions after enlarging its garbage register. Since the resource has no amplitude on the added directions, these extensions do not change F.

Support coordinates. Next we replace the uncharged garbage registers by coordinate spaces whose dimension is bounded by the footprint, using the environment of Lemma 3.2. Write Γ_{ph} for the physical frozen map of that lemma, with environment vector γ_{ph} or vectors $\gamma_{i,\mathrm{ph}}$ in $G_A G_B$; the unsubscripted Γ and γ will denote their coordinate versions.

Unitaries. Choose isometries $j_X : \mathbb{C}^K \to G_X$ whose ranges contain the local Schmidt supports of γ_{ph}, enlarging G_X if necessary. Put

$$\widehat{G}_X = X' \otimes \mathbb{C}^K, \qquad \iota_X = I_{X'} \otimes j_X : \widehat{G}_X \to X'G_X, \qquad \gamma = (j_A^\dagger \otimes j_B^\dagger)\gamma_{\mathrm{ph}}, \qquad \Gamma = E_\gamma.$$

Measurements. Let $s_i = \mathrm{SR}(\gamma_{i,\mathrm{ph}}) \ge 1$ and $S = \sum_i s_i \le K + d^2$. Choose isometries $j_{X,i} : \mathbb{C}^{s_i} \to G_X$ whose ranges contain the local Schmidt supports of $\gamma_{i,\mathrm{ph}}$, and put

$$\widehat{G}_X = \bigoplus_{i=1}^{d^2} |i\rangle \otimes \mathbb{C}^{s_i}, \qquad \iota_X(|i\rangle \otimes w) = |i\rangle_{X'} \otimes j_{X,i} w, \qquad \gamma_i = (j_{A,i}^\dagger \otimes j_{B,i}^\dagger)\gamma_{i,\mathrm{ph}},$$

and let $\Gamma = C_\gamma : \mathbb{C}^{d^2} \to \widehat{G}_A \otimes \widehat{G}_B$ map $|i\rangle$ to its two labels in $\widehat{G}_A$ and $\widehat{G}_B$ and the vector γ_i. The map ι_X is an isometry because distinct labels are orthogonal, even though the physical supports of different $\gamma_{i,\mathrm{ph}}$ need not be. In both cases $\widehat{G}_X$ replaces the output and garbage space $X'G_X$; it has dimension dK or S, and

$$(\iota_A \otimes \iota_B)\Gamma = \Gamma_{\mathrm{ph}}. \tag{18}$$

Completing the reversed decoders. Running the decoder backwards on these supports gives the contraction $W_X^{\mathrm{rev}} := W_X^\dagger \iota_X : \widehat{G}_X \to Q_X$. It fails to be an isometry exactly to the extent that the environment support leaves the range of the decoder, which can happen for an approximate protocol. We complete it to an isometry by adding a local block that records this missing amplitude:

$$J_X = \begin{pmatrix} W_X^{\mathrm{rev}} \\ Z_X \end{pmatrix} : \widehat{G}_X \to Q_X \oplus \widehat{G}_X, \qquad Z_X = (I - W_X^{\mathrm{rev}\,\dagger} W_X^{\mathrm{rev}})^{1/2}, \tag{19}$$

where the square root is the positive semidefinite one; then $J_X^\dagger J_X = W_X^{\mathrm{rev}\,\dagger} W_X^{\mathrm{rev}} + Z_X^2 = I$. In an exact protocol with minimal support coordinates, W_X^{rev} is already an isometry and $Z_X = 0$.

The two isometries. Let $\mathcal{Q} = (Q_A \oplus \widehat{G}_A) \otimes (Q_B \oplus \widehat{G}_B)$, which contains $Q_A \otimes Q_B$. The encoder isometry $\mathsf{Enc} = X_{\mathrm{ex}}(V_A \otimes V_B)E_\eta$ of [FGMS26] maps AB into $Q_A \otimes Q_B \subseteq \mathcal{Q}$. We define the decoder isometry and the overlap by

$$\mathsf{Dec} = (J_A \otimes J_B)\Gamma : \mathbb{C}^{d^2} \to \mathcal{Q}, \qquad \widehat{T} = \mathsf{Dec}^\dagger \mathsf{Enc}. \tag{20}$$

This Dec replaces the decoder isometry $(W_A \otimes W_B)^\dagger \Gamma$ of the exact proof, with which it agrees when $Z_A = Z_B = 0$. Only the blocks W_X^{rev} meet the range of Enc, so by (18)

$$\widehat{T} = \Gamma^\dagger (W_A^{\mathrm{rev}\,\dagger} \otimes W_B^{\mathrm{rev}\,\dagger}) \mathsf{Enc} = \Gamma^\dagger (\iota_A^\dagger \otimes \iota_B^\dagger)(W_A \otimes W_B)\mathsf{Enc} = \Gamma_{\mathrm{ph}}^\dagger F. \tag{21}$$

For two isometries, the squared normalized distance is determined by their overlap. Hence, with δ as in (15),

$$\|\mathsf{Enc} - \mathsf{Dec}T\|_{\mathrm{n}}^2 = 2 - \frac{2}{d^2} \operatorname{Re} \operatorname{Tr}(T^\dagger \widehat{T}) = \|F - \Gamma_{\mathrm{ph}} T\|_{\mathrm{n}}^2 \le \delta^2. \tag{22}$$

Leakage. The *leakage residuals* and the *leakage* are

$$\begin{gathered} Y_{\mathsf{Enc}} = \mathsf{Enc} - \mathsf{Dec}\widehat{T} = (I - \mathsf{Dec}\mathsf{Dec}^\dagger)\mathsf{Enc}, \qquad Y_{\mathsf{Dec}} = \mathsf{Dec} - \mathsf{Enc}\widehat{T}^\dagger = (I - \mathsf{Enc}\mathsf{Enc}^\dagger)\mathsf{Dec}, \\ h = \|Y_{\mathsf{Enc}}\|_F^2 = \|Y_{\mathsf{Dec}}\|_F^2 = d^2 - \|\widehat{T}\|_F^2. \end{gathered} \tag{23}$$

The equalities follow by expanding the squares and using $\mathsf{Enc}^\dagger \mathsf{Enc} = \mathsf{Dec}^\dagger \mathsf{Dec} = I$. Applying $\mathsf{Dec}^\dagger$ and $I - \mathsf{Dec}\mathsf{Dec}^\dagger$ to $\mathsf{Enc} - \mathsf{Dec}T$ gives

$$\|\widehat{T} - T\|_{\mathrm{n}} \le \delta, \qquad h \le d^2 \delta^2. \tag{24}$$

For an exact protocol, $\widehat{T} = T$, $h = 0$, and $\mathsf{Enc} = \mathsf{Dec}T$ is the exact factorization of [FGMS26].

4.2 The derivative identity

For unitary targets we identify the logical input and output spaces with $\mathbb{C}^d \otimes \mathbb{C}^d$ using their fixed bases. The notation $\mathfrak{u}_{\mathrm{loc}}$ denotes the corresponding copy of the local skew-Hermitian algebra on each space: a acts on the input and b on the output.

From now on we forget the protocol that produced the witness and let all blocks vary independently. The variables are the resource $\eta \in \mathbb{C}^r \otimes \mathbb{C}^r$, the encoders V_A, V_B, the completed isometries J_A, J_B, and the environment vector $\gamma \in \mathbb{C}^K \otimes \mathbb{C}^K$ or the vectors $\gamma_i \in \mathbb{C}^{s_i} \otimes \mathbb{C}^{s_i}$. They are subject only to the constraints that vectors are unit vectors and that V_A, V_B, J_A, J_B are isometries; in particular J_X is now an arbitrary isometry $\widehat{G}_X \to Q_X \oplus \widehat{G}_X$, not the completion (19) of a particular decoder. For every point of this *constraint space* we define Enc, Dec, $\widehat{T}$, Y_{Enc}, Y_{Dec} and h by the formulas above. The constraint space plays the role of the

strategy set $\mathcal{Z}_\nu$ of [FGMS26]: the isometry and normalization constraints (Z2) and (Z3) are kept, with J_X in place of W_X, while the target is no longer a coordinate, the constraint (Z4) is dropped, and the overlap $\widehat{T}$ plays the role of $U(z)$. Accuracy will be imposed later through the leakage, by the condition $h \le d^2\delta^2$. A *path* is a differentiable path in the constraint space; for an isometry block V its velocity satisfies $V^\dagger \dot{V} + \dot{V}^\dagger V = 0$.

Proposition 4.2 (Overlap derivative)**.** *Along every path in the constraint space,*

$$a := \mathsf{Enc}^\dagger \dot{\mathsf{Enc}} \in \mathfrak{u}_{\mathrm{loc}}, \qquad b := \mathsf{Dec}^\dagger \dot{\mathsf{Dec}} \in \begin{cases} \mathfrak{u}_{\mathrm{loc}} & \text{for unitaries,} \\ \mathfrak{u}_{\mathrm{diag}} & \text{for measurements,} \end{cases} \tag{25}$$

and

$$\dot{\widehat{T}} = -b\widehat{T} + \widehat{T}a + \Xi, \qquad \Xi = \dot{\mathsf{Dec}}^\dagger Y_{\mathsf{Enc}} + Y_{\mathsf{Dec}}^\dagger \dot{\mathsf{Enc}}, \qquad \|\Xi\|_F \le \sqrt{h}\,(\|\dot{\mathsf{Enc}}\|_{\mathrm{op}} + \|\dot{\mathsf{Dec}}\|_{\mathrm{op}}). \tag{26}$$

In particular, the distance of $\dot{\widehat{T}}$ from the allowed directions $\mathcal{T}(\widehat{T})$ of (4) is at most $\sqrt{h}(\|\dot{\mathsf{Enc}}\|_{\mathrm{op}} + \|\dot{\mathsf{Dec}}\|_{\mathrm{op}})$.

The locality of the generators holds at every point of the constraint space, whether or not the point comes from an accurate protocol; accuracy only enters through the leakage h. For $h = 0$, (26) is the exact identity (3).

Proof. Both Enc and Dec are isometries at every point, so a and b are anti-Hermitian. Differentiate $\widehat{T} = \mathsf{Dec}^\dagger \mathsf{Enc}$ and insert $\mathsf{Enc} = \mathsf{Dec}\widehat{T} + Y_{\mathsf{Enc}}$ in the first term and $\mathsf{Dec} = \mathsf{Enc}\widehat{T}^\dagger + Y_{\mathsf{Dec}}$ in the second:

$$\dot{\widehat{T}} = \dot{\mathsf{Dec}}^\dagger(\mathsf{Dec}\widehat{T} + Y_{\mathsf{Enc}}) + (\mathsf{Enc}\widehat{T}^\dagger + Y_{\mathsf{Dec}})^\dagger \dot{\mathsf{Enc}} = -b\widehat{T} + \widehat{T}a + \dot{\mathsf{Dec}}^\dagger Y_{\mathsf{Enc}} + Y_{\mathsf{Dec}}^\dagger \dot{\mathsf{Enc}},$$

using $\dot{\mathsf{Dec}}^\dagger \mathsf{Dec} = b^\dagger = -b$. The bound on Ξ follows from $\|Y_1 Y_2\|_F \le \|Y_1\|_{\mathrm{op}} \|Y_2\|_F$ and $\|Y_1 Y_2\|_F \le \|Y_1\|_F \|Y_2\|_{\mathrm{op}}$ together with (23).

The input generator is computed exactly as in [FGMS26]: the fixed exchange cancels, and $V_X^\dagger V_X = I$ gives

$$a = E_\eta^\dagger (V_A^\dagger \dot{V}_A \otimes I + I \otimes V_B^\dagger \dot{V}_B) E_\eta + \langle \eta \mid \dot{\eta} \rangle\, I.$$

The first two terms are local by the lemma on compression through a shared state in the preliminaries of [FGMS26], and $\langle \eta \mid \dot{\eta} \rangle$ is imaginary, so $a \in \mathfrak{u}_{\mathrm{loc}}$.

For the output generator, the product rule and $J_X^\dagger J_X = I$ give

$$b = \Gamma^\dagger (J_A^\dagger \dot{J}_A \otimes I + I \otimes J_B^\dagger \dot{J}_B) \Gamma + \Gamma^\dagger \dot{\Gamma}.$$

For unitaries, $\Gamma = E_\gamma$ inserts a state on $\mathbb{C}^K \otimes \mathbb{C}^K$, the operator $J_A^\dagger \dot{J}_A$ acts on Alice's $\widehat{G}_A = A' \otimes \mathbb{C}^K$, and the same compression lemma makes the first term local on $A'B'$; moreover $\Gamma^\dagger \dot{\Gamma} = \langle \gamma \mid \dot{\gamma} \rangle I$ is an imaginary multiple of the identity. For measurements, $\Gamma = C_\gamma$, and each entry $\langle i | C_\gamma^\dagger (J_A^\dagger \dot{J}_A \otimes I) C_\gamma | j \rangle$ contains the overlap $\langle i \mid j \rangle$ of Bob's two labels, so it vanishes for $i \ne j$; the same holds for Bob's term with Alice's labels. This is the duplicated-label lemma in the measurement appendix of [FGMS26]. Finally $C_\gamma^\dagger \dot{C}_\gamma = \mathrm{diag}(\langle \gamma_1 \mid \dot{\gamma}_1 \rangle, \dots, \langle \gamma_{d^2} \mid \dot{\gamma}_{d^2} \rangle)$. Since b is anti-Hermitian, $b \in \mathfrak{u}_{\mathrm{loc}}$ for unitaries and $b \in \mathfrak{u}_{\mathrm{diag}}$ for measurements. □

In the exact proof of [FGMS26], the output compression uses that the range of Γ lies in the range of $W_A \otimes W_B$, which follows from exactness. Here the completed maps J_X are isometries, and no such condition is needed. This is what makes the generator algebras independent of the accuracy of the protocol.

5 Haar volume and generic targets

We now count the coordinates of the witnesses, bound the derivative of the overlap map in these coordinates, and convert the bounds into a volume estimate. This proves Theorem A.

Conventions. All coordinate spaces are real Euclidean spaces: a complex entry contributes its real and imaginary parts, and matrices carry the inner product $\operatorname{Re}\operatorname{Tr}(X^\dagger Y)$. Ranks and singular values of derivatives are taken in this real-linear sense. On the target space $\mathbb{C}^{d^2\times d^2}$ we use the normalized Frobenius norm (7) unless stated otherwise. Constants $c, C, C', \dots$ are universal and may change between estimates. Within each statement they are fixed before all dimension, budget and error parameters quantified there; any other permitted dependence is stated explicitly. The constant $C_\star$ is fixed once in (27).

5.1 Coordinates and derivative bounds

Fix a factorization (r, m_A, m_B) with $r\, m_A m_B \le K$, and for measurements also the ranks $(s_i)_{i\le d^2}$ of Section 4.1. The blocks of the constraint space have the following numbers of real coordinates. The resource uses $2r^2$. The encoder V_X maps a space of dimension dr to one of dimension drm_X^2, so it uses $2d^2r^2m_X^2 \le 2d^2K^2$. For unitaries, γ uses $2K^2$, and each J_X maps $\widehat{G}_X$, of dimension dK, to $Q_X \oplus \widehat{G}_X$, of dimension at most $2dK$, so it uses at most $4d^2K^2$. For measurements, the vectors γ_i use $2\sum_i s_i^2 \le 2S^2$ and each J_X uses $2(dK+S)S$, where $S \le K + d^2$. Hence the number of coordinates is at most Cd^2K^2 for unitaries and $Cd^2(K+d^2)^2$ for measurements.

We call the set of points for one choice of these discrete data a *family*; it is the analogue of the strategy set $\mathcal{Z}_\nu$ for one dimension vector ν. There are at most K^3 factorizations, and at most $\binom{K+d^2}{d^2} \le 2^{K+d^2}$ rank allocations. We pad every family with dummy coordinates to the common number

$$D = \lceil C_\star d^2(K+d^2)^2 \rceil, \qquad \text{or} \qquad D = \lceil C_\star d^2K^2 \rceil \text{ if } K \ge d^2/4, \tag{27}$$

with a sufficiently large universal constant $C_\star \ge 16$; the second choice is possible because $K + d^2 \le 5K$ when $K \ge d^2/4$. The number of families is then at most e^{CD}.

Scaling. We rescale each block to norm one, so that the source has a universal radius. Divide V_X by $\sqrt{dr}$, its Frobenius norm. For unitaries divide J_X by $\sqrt{dK}$ and leave η and γ unchanged. For measurements divide J_X by $\sqrt{S}$, divide the tuple $(\gamma_i)_i$ by d, and leave η unchanged. There are six blocks of norm one, so the rescaled constraint space $\widetilde{\mathcal{Z}}$ lies in the ball of radius $\sqrt{6}$ in $\mathbb{R}^D$. We call

$$\widetilde{\mathcal{Z}}_\delta = \{x \in \widetilde{\mathcal{Z}} : h(x) \le d^2\delta^2\} \tag{28}$$

the *witness family*; it is compact, being a closed subset of the compact constraint space, since h is continuous. By (24), every target of error at most ε has a witness $x \in \widetilde{\mathcal{Z}}_\delta$ in one of the families, with $\|\widehat{T}(x) - T\|_{\mathrm{n}} \le \delta$.

Polynomial description. The maps Enc, Dec and $\widehat{T}$ are polynomials of fixed degree in the entries of the blocks and their complex conjugates; rescaling changes coefficients but not degrees. Each matrix constraint $V^\dagger V = I$ can be written as a single polynomial equation, the sum of the squares of the real and imaginary parts of the entries of $V^\dagger V - I$; the normalizations of the γ_i can be packaged as $\sum_i(\|\gamma_i\|^2 - 1)^2 = 0$. Together with the resource normalization, the equation setting dummy coordinates to zero, and the inequality $h \le d^2\delta^2$, these give a fixed number of polynomial equations and weak inequalities of fixed degrees defining $\widetilde{\mathcal{Z}}_\delta$ itself. The original overlap map is polynomial of degree at most six, and the leakage polynomial has degree at most twelve; these degree bounds are independent of all dimensions.

Speed. Let $\dot{x}$ be the velocity of a path in the rescaled coordinates. By the product rule, $\dot{\mathsf{Enc}}$ is a sum of three terms, containing $\dot{V}_A$, $\dot{V}_B$ and $\dot{\eta}$ respectively, and $\dot{\mathsf{Dec}}$ is a sum of three terms, containing $\dot{J}_A$, $\dot{J}_B$ and $\dot{\Gamma}$; in each term every undifferentiated factor is an isometry and has operator norm one. The derivative of the insertion map E_γ is $E_{\dot{\gamma}}$, of operator norm $\|\dot{\gamma}\|$, and the derivative of C_γ has operator norm $\max_i \|\dot{\gamma}_i\|$. We bound the operator norms of $\dot{V}_X$ and $\dot{J}_X$ by their Frobenius norms, which are $\sqrt{dr}$ and $\sqrt{dK}$ (respectively $\sqrt{S}$) times the coordinate speeds of the rescaled blocks, and the speeds of the vectors by their coordinate speeds, times d for the rescaled tuple $(\gamma_i)_i$. Cauchy–Schwarz over the six terms gives

$$\|\dot{\mathsf{Enc}}\|_{\mathrm{op}} + \|\dot{\mathsf{Dec}}\|_{\mathrm{op}} \le \beta \|\dot{x}\|, \qquad \beta^2 = \begin{cases} 2dr + 2dK + 2 & \text{for unitaries,} \\ 2dr + 2S + d^2 + 1 & \text{for measurements.} \end{cases} \tag{29}$$

The six coefficients are $1, \sqrt{dr}, \sqrt{dr}, \sqrt{dK}, \sqrt{dK}, 1$ for unitaries and $1, \sqrt{dr}, \sqrt{dr}, \sqrt{S}, \sqrt{S}, d$ for measurements.

Singular values. At $x \in \widetilde{\mathcal{Z}}_\delta$, Proposition 4.2 splits the real-linear differential of the overlap map as $\mathrm{d}\widehat{T}_x = \mathsf{L}_x + \Xi_x$, where the local part $\mathsf{L}_x(\dot{x}) = -b\widehat{T} + \widehat{T}a$ takes values in $\mathcal{T}(\widehat{T}(x))$, which has dimension at most ℓ by (5), and $\|\Xi_x(\dot{x})\|_F \le \sqrt{h}\,\beta\|\dot{x}\| \le d\delta\beta\|\dot{x}\|$. Directly from $\widehat{T} = \mathsf{Dec}^\dagger\mathsf{Enc}$ and $\|Y\|_F \le d\|Y\|_{\mathrm{op}}$ for maps with domain $\mathbb{C}^{d^2}$, also $\|\mathrm{d}\widehat{T}_x(\dot{x})\|_F \le d\beta\|\dot{x}\|$. Dividing by d to pass to the normalized norm, and using Weyl's inequality $\sigma_{\ell+1}(\mathsf{L}_x + \Xi_x) \le \sigma_{\ell+1}(\mathsf{L}_x) + \sigma_1(\Xi_x)$ with $\sigma_{\ell+1}(\mathsf{L}_x) = 0$, we obtain

$$\sigma_1(\mathrm{d}\widehat{T}_x) \le \beta \le D, \qquad \sigma_{\ell+1}(\mathrm{d}\widehat{T}_x) \le \delta\beta \le \delta D. \tag{30}$$

Here $\ell = \ell_{\mathrm{u}}$ or ℓ_{m}, and $\beta \le D$ holds for our choice of D.

Polynomial ambient extension. So far these are derivatives along the constraint space. To use an ambient Jacobian bound, we extend $\widehat{T}$ to a polynomial map on all of $\mathbb{R}^D$ while preserving its values and derivative bounds on the witness family. Before rescaling, use the cubic maps

$$R(z) = \tfrac{1}{2}(3 - \|z\|^2)z, \qquad R(V) = \tfrac{1}{2}V(3I - V^\dagger V)$$

on vector and matrix blocks, respectively; for measurements apply the vector map separately to each γ_i. They fix unit vectors and isometries, and their derivatives at such points are

$$\mathrm{d}R_z(w) = w - z\,\mathrm{Re}\langle z \mid w\rangle, \qquad \mathrm{d}R_V(Y) = Y - \tfrac{1}{2}V(V^\dagger Y + Y^\dagger V).$$

These are the orthogonal projections onto the tangent spaces of the constraints: their outputs satisfy the differentiated constraints, and the subtracted terms are orthogonal to all tangent vectors. Conjugating by each block's scalar rescaling preserves this property, and we set the dummy coordinates to zero. The cubic maps need not take nearby points into the constraint space; only their values and derivatives on it are used. Composing the original overlap polynomial with their product gives a polynomial extension of fixed degree (at most 18). At a point of $\widetilde{\mathcal{Z}}_\delta$, its derivative is the tangential differential of $\widehat{T}$ composed with an orthogonal projection, so no singular value increases and (30) holds for the ambient derivative. We continue to denote this extension by $\widehat{T}$. The witness family is still defined by the original constraints and the original leakage polynomial h; it is a compact basic closed semialgebraic set of fixed complexity.

We summarize this subsection.

Proposition 5.1 (Witness families)**.** *Let $K \ge 1$ and $0 < \varepsilon < 1$, and let δ be as in (15) and D as in (27). Every target in $\mathsf{Reach}_{K,\varepsilon}$ is within normalized distance δ of $\widehat{T}(\widetilde{\mathcal{Z}}_\delta)$ for one of at*

most e^{CD} compact families $\widetilde{\mathcal{Z}}_\delta \subseteq \mathbb{R}^D$ of radius at most $\sqrt{6}$. Each family is defined by a fixed number of polynomial equations and weak inequalities of fixed degrees, and $\widehat{T}$ has a polynomial extension of fixed degree to $\mathbb{R}^D$. On $\widetilde{\mathcal{Z}}_\delta$ its ambient derivative satisfies

$$\sigma_1(\mathrm{d}\widehat{T}) \le D, \qquad \sigma_{\ell+1}(\mathrm{d}\widehat{T}) \le \delta D,$$

with $\ell = \ell_{\mathrm{u}} = 4d^2 - 3$ for unitaries and $\ell = \ell_{\mathrm{m}} = 3d^2 - 2$ for measurements.

5.2 The transverse tube estimate

The geometric input is a quantitative version of the fact that the image of a map whose derivative has rank at most ℓ has dimension at most ℓ. We need it with constants that are exponential in the number of variables and otherwise independent of all dimensions.

Definition 5.2 (Bounded polynomial format)**.** *Let $Y \subseteq \mathbb{R}^k$ and let $\phi : \mathbb{R}^k \to \mathbb{R}^m$ be polynomial. We say that (Y, ϕ) has* bounded polynomial format *if Y is defined by a conjunction of at most Δ_0 polynomial equations and weak inequalities, all of degree at most Δ_0, and every coordinate of ϕ has degree at most Δ_0. Here $\Delta_0 \ge 2$ is one absolute constant, chosen large enough for the finitely many kinds of maps in these notes, independently of all dimensions. We write C_Δ for constants that depend only on Δ_0 and on a bound for the radius of Y.*

By Proposition 5.1, the witness families and their polynomial overlap extensions have bounded polynomial format. The same is true after adjoining a Euclidean ball and adding its coordinates to the output, or after composing the overlap with the degree-four invariant used in Section 8. Increasing the fixed budget Δ_0 accommodates these operations.

Lemma 5.3 (Polynomial image volume)**.** *Let $k, m \ge 1$, let $Y \subset \mathbb{R}^k$ be compact and contained in a ball of radius R_0, and let $\phi : \mathbb{R}^k \to \mathbb{R}^m$ be polynomial with (Y, ϕ) of bounded polynomial format. Then*

$$\mathrm{Vol}_m(\phi(Y)) \le C_\Delta^{k+m}\, \omega_m \sup_{y \in Y} J_m\phi(y),$$

where ω_m is the volume of the unit ball in $\mathbb{R}^m$ and $J_m\phi(y)$ is the product of the m largest singular values of the derivative (zero if $k < m$). The empty image has volume zero.

The proof in Appendix A selects a maximizer of a linear function on each regular fibre, bounds the coordinate fibres of the resulting Lagrange-multiplier sets by an elementary polynomial zero count, and applies the area formula. Applying the lemma to the thickened map $(x, Y') \mapsto \widehat{T}(x) + sY'$ gives the following estimate.

Proposition 5.4 (Transverse tube estimate)**.** *Let $\widetilde{\mathcal{Z}} \subset \mathbb{R}^D$ be compact and contained in a ball of universal radius, and let $\widehat{T} : \mathbb{R}^D \to \mathbb{C}^{d^2 \times d^2}$, with $d \ge 2$ and $N = d^4$, be polynomial such that $(\widetilde{\mathcal{Z}}, \widehat{T})$ has bounded polynomial format. Suppose that, in normalized Frobenius norm,*

$$\sigma_1(\mathrm{d}\widehat{T}_x) \le L, \qquad \sigma_{\ell+1}(\mathrm{d}\widehat{T}_x) \le \lambda \qquad (x \in \widetilde{\mathcal{Z}})$$

for some $0 \le \ell \le N$. Let $\mathcal{S} \subseteq \mathrm{U}(d^2)$ be measurable with $\mathrm{dist}_{\mathrm{n}}(T, \widehat{T}(\widetilde{\mathcal{Z}})) \le \tau$ for all $T \in \mathcal{S}$. If $\tau + \lambda \le s \le c_0$ with $s > 0$, then

$$\mu(\mathcal{S}) \le e^{C_\Delta(D+N)} \left[C_\Delta(L+s)\right]^\ell \left(C_\Delta s\right)^{N-\ell},$$

where μ is the Haar probability measure and one can take $c_0 = 1/64$. A union of at most e^{CD} such sets only changes the constant.

The mechanism is as follows. Adding a matrix of normalized Frobenius norm at most s to $\widehat{T}$ produces an ambient Jacobian with at most ℓ large factors, of size $L+s$, and $2N-\ell$ small factors, of size at most $\lambda+s \le 2s$. Because $\mathrm{U}(d^2)$ has N normal directions, the s-neighbourhood of $\mathcal{S}$ in the $2N$-dimensional matrix space has volume at least of order $s^N \mu(\mathcal{S})$, in normalized units; only this lower bound is needed, and Lemma A.2 gives the exact factors. Comparing the two leaves $N-\ell$ small factors. The proof in Appendix A keeps track of the natural scale d of the unitary group in Frobenius norm, so that the threshold c_0 does not depend on d.

For measurements we work with the matrices $T = M^\dagger$. A protocol that witnesses the basis $(\varphi_i)_i$ also witnesses $(e^{i\theta_i}\varphi_i)_i$, after replacing γ_i by $e^{i\theta_i}\gamma_i$, so the sets $\mathsf{Reach}^{\mathrm{m}}_{K,\varepsilon}$ consist of whole fibres $\{\Delta M^\dagger\}$. The diagonal generators $b \in \mathfrak{u}_{\mathrm{diag}}$ in $\mathcal{T}_{\mathrm{m}}$ account for exactly these directions.

5.3 The precision bound for generic targets

Theorem 5.5 (Haar volume at fixed footprint)**.** *There is a universal constant C such that for every $d \ge 2$, $K \ge d^2/4$ and $0 < \varepsilon \le 1/2$,*

$$\mu(\mathsf{Reach}^{\mathrm{u}}_{K,\varepsilon}) \le \min\{1,\ e^{Cd^2K^2}\,\varepsilon^{\kappa_{\mathrm{u}}/2}\}, \qquad \mu(\mathsf{Reach}^{\mathrm{m}}_{K,\varepsilon}) \le \min\{1,\ e^{Cd^2K^2}\,\varepsilon^{\kappa_{\mathrm{m}}/2}\}, \tag{31}$$

with the codimensions $\kappa_{\mathrm{u}} = (d^2-1)(d^2-3)$ and $\kappa_{\mathrm{m}} = (d^2-1)(d^2-2)$ of (6). For measurements, $\mu(\mathsf{Reach}^{\mathrm{m}}_{K,\varepsilon})$ is the invariant measure of the set of approximable PVMs.

Proof. Since $K \ge d^2/4$ we can use $D = \lceil C_\star d^2K^2 \rceil$ for both target classes. Put $\tau = 2\sqrt{\varepsilon}$, which is at least δ in both cases, so every target in $\mathsf{Reach}_{K,\varepsilon}$ is within normalized distance τ of $\widehat{T}(\widetilde{\mathcal{Z}}_\tau)$ for one of e^{CD} families. Apply Proposition 5.4 to the set $\mathcal{S}$ of targets within distance τ of $\widehat{T}(\widetilde{\mathcal{Z}}_\tau)$, with the bounds of Proposition 5.1:

$$L = D, \qquad \lambda = \tau D, \qquad s = \tau + \lambda \le 2D\tau = 4D\sqrt{\varepsilon}.$$

If $s \le c_0$, the proposition gives

$$\mu(\mathsf{Reach}_{K,\varepsilon}) \le e^{C(D+N)}(CD)^{\ell}(CD\sqrt{\varepsilon})^{N-\ell} \le e^{C(D+N)}(CD)^N \varepsilon^{\kappa/2}, \tag{32}$$

with ℓ and $\kappa = N-\ell$ for the relevant class. Since $K \ge d^2/4$ and $C_\star \ge 16$, we have $D \ge d^6$, hence $N = d^4 \le D^{2/3}$ and $N\ln(CD) \le C'D$. All factors other than $\varepsilon^{\kappa/2}$ are therefore at most $e^{CD} \le e^{C'd^2K^2}$, which proves (31) in this case.

If $s > c_0$, then $\varepsilon > c_0^2/(16D^2)$, so $\ln(1/\varepsilon) \le C(1+\ln D)$ and $\frac{\kappa}{2}\ln(1/\varepsilon) \le CN(1+\ln D) \le C'D$. After enlarging C, the right-hand side of (31) is then at least one. For measurements, the reachable set consists of whole diagonal fibres, so its Haar measure is the invariant measure of its image in $\mathrm{U}(d^2)/\mathrm{U}(1)^{d^2}$. □

The factor $\varepsilon^{\kappa/2} = (\sqrt{\varepsilon})^\kappa$ agrees with the volume of a tube of radius $\sqrt{\varepsilon}$ around an orbit of codimension κ. The proof does not show that approximable targets are close to exactly implementable ones; it controls the images of the low-leakage witnesses directly. The prefactor $e^{Cd^2K^2}$ accounts for the $D = O(d^2K^2)$ parameters.

Corollary 5.6 (Universal approximation)**.** *There are universal constants $c, C > 0$ such that, for $d \ge 2$ and $0 < \varepsilon \le 1/2$, a footprint K that suffices for every unitary with $\delta_{\mathrm{Choi}} \le \varepsilon$, or for every rank-one PVM with $\delta_\Phi \le \varepsilon$, satisfies*

$$K \ge c\,d\sqrt{\ln(1/\varepsilon)}, \qquad \textit{equivalently} \qquad \varepsilon \ge \exp(-CK^2/d^2).$$

For $d = 2^n$ this reads $\log_2 K \ge n + \frac{1}{2}\log_2\ln(1/\varepsilon) - O(1)$.

Proof. The Schmidt-rank floors of Lemma B.1 give $K \geq d^2(1-\varepsilon) \geq d^2/2$ for all unitaries and $K \geq d^2(1-\varepsilon)^2 \geq d^2/4$ for all PVMs, so Theorem 5.5 applies. Universality makes the reachable set the whole group, of measure one. Taking logarithms in (31) gives $\frac{\kappa}{2}\ln(1/\varepsilon) \leq Cd^2K^2$, and $\kappa \geq \frac{3}{16}d^4$ by (6). □

Corollary 5.7 (Almost every fixed target)**.** *There is a universal $c > 0$ such that, for each $d \geq 2$, Haar-almost every $U \in \mathrm{U}(d^2)$ has a threshold $\varepsilon_0(U) > 0$ with*

$$K_\varepsilon(U) \geq c\, d\sqrt{\ln(1/\varepsilon)} \qquad (0 < \varepsilon < \varepsilon_0(U)).$$

The same holds for almost every rank-one PVM, with respect to the invariant measure on $\mathrm{U}(d^2)/\mathrm{U}(1)^{d^2}$.

Proof. Let $\varepsilon_j = e^{-j}$ and $K_j = \lfloor c_1 d\sqrt{j} \rfloor$ with $c_1 > 0$ small. For all large j we have $K_j \geq d^2$, and Theorem 5.5 gives

$$\mu(\mathsf{Reach}_{K_j,\varepsilon_j}) \leq \exp(Cc_1^2 d^4 j - \tfrac{3}{32}d^4 j) \leq e^{-c_2 d^4 j},$$

which is summable in j. By the Borel–Cantelli lemma, almost every target lies in only finitely many of the sets $\mathsf{Reach}_{K_j,\varepsilon_j}$, so $K_{\varepsilon_j}(U) > K_j$ for all $j \geq j_0(U)$. For $\varepsilon_{j+1} \leq \varepsilon < \varepsilon_j$, monotonicity gives $K_\varepsilon(U) \geq K_{\varepsilon_j}(U) > K_j \geq c\, d\sqrt{\ln(1/\varepsilon)}$ after decreasing c. □

Remark 5.8 (Parameters against codimension). The witness families have $D = O(d^2K^2)$ parameters, while the target directions transverse to local equivalence number $\kappa = \Theta(d^4)$. The tube estimate compares $\ln(1/\varepsilon)$ with $D/\kappa = O(K^2/d^2)$, which is the origin of the factor d in Corollaries 5.6 and 5.7. The quantum calculation and this comparison are the same for unitaries and measurements. In the next section we reduce D for targets near SWAP and near a basis of maximally entangled vectors.

6 Universal approximation: targets near SWAP and near a maximally entangled basis

The bound of Corollary 5.6 uses the same witness construction for every target. For universal approximation we can do better by testing the protocol on a fixed ball of targets: for unitaries a ball around SWAP, the unitary that sends Alice's input to Bob's output and vice versa, and for measurements a ball around a basis of maximally entangled vectors. Such targets use both communication directions fully: at small error both messages must have dimension of order d, which leaves Schmidt rank $O(K/d^2)$ for the resource, and the flat Schmidt spectra of these targets let the environment be truncated to total Schmidt rank $O(K/d^2)$ near SWAP and $O(K/d)$ near the maximally entangled basis. This reduces the number of parameters from $O(d^2K^2)$ to $O(K^2)$. Since we only need to control a constant fraction of the target directions, a pointwise cutoff on the singular values of the leakage residuals suffices for the derivative estimate.

Throughout this section $N = d^4$. In the normalized Frobenius norm, $\mathrm{U}(d^2)$ lies in the unit sphere of $\mathbb{R}^{2N}$. Hence, for every $W \in \mathrm{U}(d^2)$ and $0 < t \leq 1$,

$$\mu\{V \in \mathrm{U}(d^2) : \|V - W\|_{\mathrm{n}} \leq t\} \geq (1 + 2/t)^{-2N}. \tag{33}$$

Indeed, a maximal t-separated subset of $\mathrm{U}(d^2)$ has at most $(1 + 2/t)^{2N}$ points, because the balls of radius $t/2$ around its points are disjoint and lie in the ball of radius $1 + t/2$; the balls of radius t around these points cover $\mathrm{U}(d^2)$, and by invariance of μ and of the norm under left multiplication they all have the same Haar measure as the ball around W.

6.1 Unitaries

Let

$$\mathcal{B}_d = \{U \in \mathrm{U}(d^2) : \|U - \mathsf{SWAP}\|_{\mathrm{n}} \le 1/4\}.$$

By (33) with $t = \frac{1}{4}$,

$$\mu(\mathcal{B}_d) \ge 9^{-2N}. \tag{34}$$

Lemma 6.1 (Compression near SWAP)**.** *Let $U \in \mathcal{B}_d$ have a protocol of footprint at most K with $\delta_{\mathrm{Choi}} \le \varepsilon \le 1/16$, and select a pure resource component with score at least $1-\varepsilon$, of Schmidt rank r. Then*

$$m_A,\, m_B \ge d/\sqrt{2}, \qquad K \ge d^2/2, \qquad r \le 2K/d^2.$$

Moreover, there is a unit vector $\gamma \in G_A G_B$ with

$$\mathrm{SR}(\gamma) \le k_\gamma := \lceil K/\lceil d^2/2\rceil\rceil \le 4K/d^2, \qquad \|F - E_\gamma U\|_{\mathrm{n}} \le \sqrt{21\varepsilon/2}. \tag{35}$$

With these bounds, the witness construction of Section 4.1 uses $D = O(K^2)$ real coordinates.

Proof. Messages. Consider the protocol on the Choi input, and let ρ be the state of Alice's reference $\bar{A}$ together with Bob's kept register K_B after the encoders. Bob's encoder does not touch A or $\bar{A}$, so $\rho = I/d \otimes \sigma$ for a state σ on K_B. For every state on a product $\mathcal{H}_1 \otimes \mathcal{H}_2$ we have $\rho_{12} \le (\dim \mathcal{H}_2)\, \rho_1 \otimes I_2$: for a pure state $\sum_j |\psi_j\rangle|j\rangle$ this is Cauchy–Schwarz in the form $|\sum_j \langle\phi_j \mid \psi_j\rangle|^2 \le (\dim \mathcal{H}_2) \sum_j |\langle\phi_j \mid \psi_j\rangle|^2$, and mixed states follow by convexity. Applying this to Bob's incoming message $M_{B_{\mathrm{post}}}$, of dimension m_A, and then Bob's decoding channel gives, for the state of $\bar{A}B'$,

$$\rho_{\bar{A}B'} \le m_A^2\, (I/d \otimes \omega)$$

for some state ω on B'; the second factor m_A comes from writing $I_{M_{B_{\mathrm{post}}}} = m_A \cdot I_{M_{B_{\mathrm{post}}}}/m_A$. Hence the fidelity of $\rho_{\bar{A}B'}$ with the maximally entangled state of $\bar{A}$ and B' is at most m_A^2/d^2. For SWAP this fidelity equals one. The Choi vector of U is within distance $\frac{1}{4}$ of that of SWAP, and the Choi state of the protocol is within trace distance $\sqrt{\varepsilon} \le \frac{1}{4}$ of that of U, by the Fuchs–van de Graaf inequality. Hence $m_A^2/d^2 \ge \frac{1}{2}$. The same argument applies to m_B, and then $K \ge r\, m_A m_B \ge r d^2/2$.

Environment. Let $\varpi_1 \ge \varpi_2 \ge \dots$ be the squared Schmidt coefficients of $|U\rangle\!\rangle$ across the laboratory cut. The Choi vector of SWAP is a product of two maximally entangled states, of $\bar{A}$ with B' and of $\bar{B}$ with A', so it has d^2 Schmidt coefficients equal to $1/d$. Mirsky's inequality for singular values [Bha97] gives $\sum_i (\sqrt{\varpi_i} - d^{-1})^2 \le \|U - \mathsf{SWAP}\|_{\mathrm{n}}^2 \le \frac{1}{16}$. Each index with $\varpi_i < 1/(4d^2)$ contributes more than $1/(4d^2)$ to this sum, so at least $3d^2/4$ indices have $\varpi_i \ge 1/(4d^2)$. Select $k_0 = \lceil d^2/2\rceil$ of them and put $\alpha_0 = k_0/(4d^2) \ge \frac{1}{8}$. Then $\varpi \ge \alpha_0 \bar{u}$ entrywise, where $\bar{u}$ is the uniform distribution on the selected indices. For every probability vector λ, comparing the mass omitted by the best K products gives

$$1 - \Lambda_K(\varpi \otimes \lambda) \ge \alpha_0 \big[1 - \Lambda_K(\bar{u} \otimes \lambda)\big] \ge \alpha_0 \big[1 - \Lambda_{\lceil K/k_0\rceil}(\lambda)\big], \tag{36}$$

where Λ_k denotes the sum of the k largest weights; the second inequality holds because each λ_j/k_0 occurs k_0 times in $\bar{u} \otimes \lambda$.

Now let γ' and the score $p = q_U \ge 1 - \varepsilon$ be as in the proof of Lemma 3.2(i), and let λ be the Schmidt spectrum of γ'. The Choi vector of F has Schmidt rank at most K and overlap $\sqrt{p}$ with $|U\rangle\!\rangle \otimes \gamma'$, whose spectrum is $\varpi \otimes \lambda$, so $\Lambda_K(\varpi \otimes \lambda) \ge p$. By (36), the top-$k_\gamma$ truncation of γ' retains mass $t \ge 1 - 8\varepsilon$. Let γ be this truncation, normalized. As in Lemma 3.2, the overlap of $|F\rangle\!\rangle$ with $|U\rangle\!\rangle \otimes \gamma$ is $\sqrt{pt}$. For $0 \le \varepsilon \le 1/16$,

$$(1-\varepsilon)(1-8\varepsilon) - \big(1 - \tfrac{21}{4}\varepsilon\big)^2 = \varepsilon\big(\tfrac{3}{2} - \tfrac{313}{16}\varepsilon\big) \ge 0.$$

Since $1 - \frac{21}{4}\varepsilon \geq 0$, we obtain $\sqrt{pt} \geq 1 - \frac{21}{4}\varepsilon$ and hence $\|F - E_\gamma U\|_{\mathrm{n}}^2 \leq \frac{21}{2}\varepsilon$. Finally $k_\gamma \leq K/\lceil d^2/2 \rceil + 1 \leq 2K/d^2 + 1 \leq 4K/d^2$, since $K \geq d^2/2$.

Coordinates. Since $r\, m_X \leq K/m_{X^c} \leq \sqrt{2}K/d$, where X^c is the other party, each encoder uses $O(d^2 r^2 m_X^2) = O(K^2)$ coordinates. The support spaces are $\widehat{G}_X = X' \otimes \mathbb{C}^{k_\gamma}$, of dimension dk_γ, and Q_X has dimension at most dK, so each completed isometry J_X uses $O(d^2 K k_\gamma + d^2 k_\gamma^2) = O(K^2)$ coordinates. The resource and γ use $O(r^2 + k_\gamma^2) = O(K^2)$. □

Lemma 6.2 (Most target directions move slowly)**.** *The witnesses of Lemma 6.1, with $\delta = \sqrt{21\varepsilon/2}$, admit coordinates of universal radius and ambient extensions as in Section 5.1 for which, in normalized Frobenius norm,*

$$\sigma_1(\mathrm{d}\widehat{T}) \leq C\sqrt{K/d^2}, \qquad \sigma_{\ell_\star+1}(\mathrm{d}\widehat{T}) \leq C\delta\sqrt{K/d^2}, \qquad \ell_\star := 4d^2 - 3 + 4d^2\lfloor d^2/64 \rfloor \leq \tfrac{7}{8}N. \tag{37}$$

Proof. Divide each isometry block with k columns by $\sqrt{k}$ and leave the vectors unchanged. We now bound Frobenius norms of the velocities of Enc and Dec, instead of operator norms. For the encoders, the compression lemma of [FGMS26] gives

$$\|(\dot{V}_A \otimes V_B)E_\eta\|_F^2 = d\ \mathrm{Tr}[\dot{V}_A{}^\dagger \dot{V}_A\, (I_d \otimes \eta_A)] \leq d\,\|\dot{V}_A\|_F^2,$$

where η_A is the reduced state of the resource. In the scaled coordinates $\dot{V}_A$ is $\sqrt{dr}$ times the coordinate velocity, so this term contributes at most $d\sqrt{r}$ times the coordinate speed. In the same way the completed isometries contribute at most $d\sqrt{k_\gamma}$, and the two vectors at most d. Since $r, k_\gamma \leq 4K/d^2$ and $d^2 \leq 2K$,

$$\|\mathrm{d}\mathsf{Enc}(v)\|_F + \|\mathrm{d}\mathsf{Dec}(v)\|_F \leq C\sqrt{K}\,\|v\|. \tag{38}$$

The same bound holds for $\mathrm{d}\widehat{T}(v) = \mathrm{d}\mathsf{Dec}(v)^\dagger \mathsf{Enc} + \mathsf{Dec}^\dagger \mathrm{d}\mathsf{Enc}(v)$, which gives the first bound in (37) after dividing by d.

Both leakage residuals have $\|Y_{\mathsf{Enc}}\|_F^2 = \|Y_{\mathsf{Dec}}\|_F^2 = h \leq d^2\delta^2$. Split each of them at singular value 8δ into a low part, of operator norm at most 8δ, and a high part, of rank at most $\lfloor d^2/64 \rfloor$. In the residual velocity $\Xi = \mathrm{d}\mathsf{Dec}(v)^\dagger Y_{\mathsf{Enc}} + Y_{\mathsf{Dec}}^\dagger \mathrm{d}\mathsf{Enc}(v)$, the high part of Y_{Enc} has a right support of dimension at most $\lfloor d^2/64 \rfloor$ that does not depend on v, and the high part of $Y_{\mathsf{Dec}}^\dagger$ has a left support of the same dimension. As real-linear maps of v, the two high contributions therefore take values in a fixed space of real dimension at most $4d^2\lfloor d^2/64 \rfloor$. Together with the allowed directions, of dimension at most $4d^2 - 3$, this gives a space of dimension at most $\ell_\star$. Let Π be the orthogonal projection onto this space. Then $\Pi\,\mathrm{d}\widehat{T}$ has rank at most $\ell_\star$, and $(I - \Pi)\,\mathrm{d}\widehat{T}(v)$ is the projection of the low contributions, of Frobenius norm at most $8\delta(\|\mathrm{d}\mathsf{Dec}(v)\|_F + \|\mathrm{d}\mathsf{Enc}(v)\|_F) \leq C\delta\sqrt{K}\|v\|$ by (38). Dividing by d and using $\sigma_{\ell_\star+1}(\mathrm{d}\widehat{T}) \leq \|\mathrm{d}\widehat{T} - \Pi\,\mathrm{d}\widehat{T}\|_{\mathrm{op}}$, which holds because $\Pi\,\mathrm{d}\widehat{T}$ has rank at most $\ell_\star$, proves the second bound. The split is pointwise bookkeeping and need not vary smoothly. The ambient extension of Section 5.1 precomposes the derivative with an orthogonal projection, so the bounds survive. Finally, $(4d^2 - 3)/d^4 \leq 13/16$ and $4d^2\lfloor d^2/64 \rfloor \leq N/16$ for $d \geq 2$. □

Theorem 6.3 (Haar volume near SWAP)**.** *There is a universal C such that for $d \geq 2$, $K \geq d^2/2$ and $0 < \varepsilon \leq 1/2$,*

$$\mu(\mathcal{B}_d \cap \mathsf{Reach}^{\mathrm{u}}_{K,\varepsilon}) \leq \min\{1,\ e^{CK^2}\,\varepsilon^{d^4/16}\}.$$

Proof. First let $\varepsilon \leq 1/16$, so $\delta = \sqrt{21\varepsilon/2} \leq 1$. The witness families of Lemma 6.1 have $D = O(K^2)$ coordinates, bounded polynomial format, and at most e^{CK^2} members, and every target is within normalized distance δ of an image. Put $u = K/d^2 \geq \frac{1}{2}$ and apply Proposition 5.4

with the bounds of Lemma 6.2, with $\ell = \ell_\star$, $\tau = \delta$, and $s = C_2\delta\sqrt{1+u}$ for a constant C_2 large enough that $s \geq \tau + \lambda$. If $s \leq c_0$, then

$$\mu(\mathcal{B}_d \cap \mathsf{Reach}^{\mathrm{u}}_{K,\varepsilon}) \leq e^{C(K^2+N)}\,[C\sqrt{1+u}]^N\,\delta^{N-\ell_\star} \leq e^{C'K^2}\varepsilon^{N/16},$$

because $N - \ell_\star \geq N/8$, $\delta^{N/8} = (21\varepsilon/2)^{N/16}$, $N \leq 4K^2$, and $N\ln(1+u) \leq Nu \leq 2K^2$. If $s > c_0$, then $\ln(1/\varepsilon) \leq C + \ln(1+u)$, and the claimed bound is at least one after enlarging C. The same holds for $1/16 < \varepsilon \leq 1/2$. □

The exponent $N/16$ is smaller than the exponent $\kappa_{\mathrm{u}}/2 \approx N/2$ of Theorem 5.5; the gain is in the prefactor, e^{CK^2} in place of $e^{Cd^2K^2}$, and it comes from the smaller witness families.

Corollary 6.4 (Universal unitary approximation)**.** *For $d \geq 2$ and $0 < \varepsilon \leq 1/2$, a footprint K that suffices for every unitary with $\delta_{\mathrm{Choi}} \leq \varepsilon$ satisfies*

$$K \geq d^2 \max\left\{1-\varepsilon,\ \frac{\sqrt{\ln(1/\varepsilon)}}{51}\right\}.$$

For $d \geq 8$, the denominator 51 can be replaced by 24. For $d = 2^n$, this gives $\log_2 K \geq 2n + \frac{1}{2}\log_2\ln(1/\varepsilon) - 6$, with 6 replaced by 5 when $n \geq 3$.

Proof. The Schmidt-rank floor for SWAP (Lemma B.1) gives $K \geq d^2(1-\varepsilon) \geq d^2/2$. Universality puts $\mathcal{B}_d$ inside the reachable set, so (34) and Theorem 6.3 give $9^{-2N} \leq e^{CK^2}\varepsilon^{N/16}$. Taking logarithms gives the precision term up to a universal constant. Keeping the numerical factors in the volume estimate gives the constants 51 and 24, as detailed in Appendix D. □

6.2 Measurements

For measurements we replace SWAP by a basis of maximally entangled vectors. Let $|\Phi^+\rangle = d^{-1/2}\sum_a |a\rangle|a\rangle$, let $\mathsf{W}_1, \ldots, \mathsf{W}_{d^2}$ be the Weyl operators (generalized Pauli operators) on $\mathbb{C}^d$, and let $M_{\mathrm{Bell}} = \sum_i |e_i\rangle\langle i|$ be the basis matrix of the maximally entangled basis $e_i = (I \otimes \mathsf{W}_i)|\Phi^+\rangle$. Put

$$t_d = \frac{1}{4d^{3/2}}, \qquad \mathcal{M}_d = \{M \in \mathrm{U}(d^2) : \|M - M_{\mathrm{Bell}}\|_{\mathrm{n}} \leq t_d\}, \qquad \mathcal{M}_d^\dagger = \{M^\dagger : M \in \mathcal{M}_d\}.$$

The ball is much smaller than $\mathcal{B}_d$, but by (33) and the inversion invariance of μ,

$$\mu(\mathcal{M}_d^\dagger) = \mu(\mathcal{M}_d) \geq (1 + 8d^{3/2})^{-2N}, \tag{39}$$

which is still $e^{-O(N\ln d)}$. Its radius is chosen so that every basis vector in $\mathcal{M}_d$ has an almost flat Schmidt spectrum.

Lemma 6.5 (Compression near a maximally entangled basis)**.** *Let $M \in \mathcal{M}_d$ have a protocol of footprint at most K with $\delta_\Phi \leq \varepsilon \leq 1/256$, and select a pure resource component with score at least $1-\varepsilon$, of Schmidt rank r. Then*

$$m_A,\, m_B \geq d/\sqrt{2}, \qquad K \geq d^3/2, \qquad r \leq 2K/d^2.$$

Moreover, there are unit vectors $\gamma_1, \ldots, \gamma_{d^2} \in G_A G_B$ with

$$\sum_i \mathrm{SR}(\gamma_i) \leq \lceil K/d \rceil + d^2 \leq 4K/d, \qquad \|F - C_\gamma M^\dagger\|_{\mathrm{n}} \leq \sqrt{50\varepsilon/9}. \tag{40}$$

With these bounds, the witness construction of Section 4.1 uses $D = O(K^2)$ real coordinates.

Proof. Flat spectra. Let φ_i be the columns of M. Then $\|\varphi_i - e_i\| \le \|M - M_{\mathrm{Bell}}\|_F \le dt_d = \frac{1}{4}d^{-1/2}$. Reshaping a vector of $\mathbb{C}^d \otimes \mathbb{C}^d$ into its $d \times d$ coefficient matrix preserves the Euclidean norm, the coefficient matrix of e_i has all singular values equal to $d^{-1/2}$, and singular values are 1-Lipschitz in the Frobenius norm [Bha97]. Hence all d Schmidt coefficients of φ_i lie in $[\frac{3}{4}d^{-1/2}, \frac{5}{4}d^{-1/2}]$, and the squared ones satisfy

$$\frac{9}{16d} \le \lambda_{ij} \le \frac{25}{16d} \qquad (1 \le j \le d). \tag{41}$$

Footprint. We use the notation of the proof of Lemma 3.2(ii), with $p = q_\Phi \ge 1 - \varepsilon$. Since $\zeta_{ik}^2 \le \|\tilde{\gamma}_i\|^2 \le 1$, every squared Schmidt coefficient (17) of v_0 is at most $25/(16d^3)$, so $p^2 \le \Lambda_K(v_0) \le 25K/(16d^3)$ and $K \ge \frac{16}{25}(1-\varepsilon)^2 d^3 \ge d^3/2$.

Messages. Run the protocol on the Choi input, in which A is maximally entangled with Alice's reference $\bar{A}$ and B with Bob's reference $\bar{B}$. For the frozen dilation $C_{\gamma'}M^\dagger$ of Lemma 3.2(ii), both parties output the same label i, with probability d^{-2}, and the references are then left in the state $|\bar{\varphi}_i\rangle$, since $|\Omega\rangle = d^{-1}\sum_i |\varphi_i\rangle|\bar{\varphi}_i\rangle$. Let Bob, who holds $\bar{B}$ and his label i, apply $\bar{\mathsf{W}}_i^\dagger$ to $\bar{B}$. Since $|\bar{e}_i\rangle = (I \otimes \bar{\mathsf{W}}_i)|\Phi^+\rangle$, the pair $\bar{A}\bar{B}$ then has fidelity $|\langle e_i \mid \varphi_i\rangle|^2 \ge 1 - \|\varphi_i - e_i\|^2$ with $|\Phi^+\rangle$, and on average over i at least $1 - t_d^2$. For the actual protocol F, whose Choi vector is within $\|F - C_{\gamma'}M^\dagger\|_{\mathrm{n}} \le 2\sqrt{\varepsilon}$ of that of $C_{\gamma'}M^\dagger$, the same local correction by Bob gives fidelity at least $1 - t_d^2 - 2\sqrt{\varepsilon} \ge \frac{1}{2}$, because trace distance between pure states is at most their Euclidean distance and channels do not increase it. On the other hand, before Bob receives Alice's message, the state of $\bar{A}$ together with Bob's kept register and $\bar{B}$ is $I/d \otimes \sigma$, because $\bar{A}$ is correlated only with Alice's input and Alice's operations do not change this reduced state. Exactly as in the proof of Lemma 6.1, with $\bar{B}$ among Bob's registers, the final state of $\bar{A}\bar{B}$ is bounded by $m_A^2(I/d \otimes \omega)$ for a state ω, so its fidelity with $|\Phi^+\rangle$ is at most m_A^2/d^2. Hence $m_A^2 \ge d^2/2$. The same argument with the roles of Alice and Bob exchanged, Alice correcting $\bar{A}$ with her label, gives $m_B^2 \ge d^2/2$, and then $r \le K/(m_A m_B) \le 2K/d^2$.

Environment. Let $a_{ik} = d^{-2}\zeta_{ik}^2$, so that $\sum_{i,k} a_{ik} = p$ and the squared Schmidt coefficients of v_0 are $w_{ijk} = a_{ik}\lambda_{ij}$. For a nonnegative vector x let $\Lambda_k(x)$ be the sum of its k largest entries. If a set $\mathcal{E}$ of K triples (i,j,k) carries the top-K mass of v_0, then, by (41),

$$p - \Lambda_K(v_0) = \sum_{(i,j,k)\notin\mathcal{E}} w_{ijk} \ge \frac{9}{16} \sum_{(i,j,k)\notin\mathcal{E}} \frac{a_{ik}}{d} \ge \frac{9}{16}\Big[p - \Lambda_{\lceil K/d\rceil}(a)\Big],$$

where the last step holds because each pair (i,k) occurs in at most d triples, so the K triples of $\mathcal{E}$ carry at most the mass $\Lambda_{\lceil K/d\rceil}(a)$ of the vector $(a_{ik}/d)_{i,j,k}$. With $\Lambda_K(v_0) \ge p^2$ this gives $p - \Lambda_{\lceil K/d\rceil}(a) \le \frac{16}{9}(p - p^2) \le \frac{16}{9}\varepsilon$. Now keep in each $\tilde{\gamma}_i$ only the Schmidt terms k for which a_{ik} is among the $\lceil K/d\rceil$ largest entries of a, and define t_i and γ_i from these terms as in the proof of Lemma 3.2(ii). Then $\sum_i \mathrm{SR}(\gamma_i) \le \lceil K/d\rceil + d^2$ and $d^{-2}\sum_i t_i^2 = \Lambda_{\lceil K/d\rceil}(a) \ge p - \frac{16}{9}\varepsilon \ge 1 - \frac{25}{9}\varepsilon$, so the overlap estimate of that proof gives $\|F - C_\gamma M^\dagger\|_{\mathrm{n}}^2 \le \frac{50}{9}\varepsilon$. Since $K \ge d^3/2$, we have $d^2 \le 2K/d$ and $1 \le K/d$, so $\lceil K/d\rceil + d^2 \le 4K/d$.

Coordinates. Since $r\, m_X \le K/m_{X^c} \le \sqrt{2}K/d$, where X^c is the other party, each encoder uses $O(d^2 r^2 m_X^2) = O(K^2)$ coordinates, and the resource uses $O(r^2) = O(K^2)$. With $S = \sum_i \mathrm{SR}(\gamma_i) \le 4K/d$, the vectors γ_i use $O(S^2)$ coordinates and each completed isometry J_X, from $\widehat{G}_X$ of dimension S to $Q_X \oplus \widehat{G}_X$ of dimension at most $dK + S$, uses $O((dK+S)S) = O(K^2)$ coordinates. There are at most K^3 factorizations and at most $\binom{\lceil K/d\rceil + d^2}{d^2} \le 2^{\lceil K/d\rceil + d^2} \le 2^{3K}$ rank allocations. □

Lemma 6.6 (Most target directions move slowly)**.** *The witnesses of Lemma 6.5, with $\delta = \sqrt{50\varepsilon/9}$, admit coordinates of universal radius and ambient extensions as in Section 5.1 for which the bounds (37) hold, with the same $\ell_\star$.*

Proof. Use the scaling of Section 5.1 for measurements. As in the proof of Lemma 6.2, the encoders contribute at most $d\sqrt{r}$ times the coordinate speed to $\|\mathrm{d}\mathsf{Enc}(v)\|_F$, and the resource at most d. For the completed isometries,

$$\|(\mathrm{d}J_A \otimes J_B)C_\gamma\|_F^2 = \mathrm{Tr}\big[\mathrm{d}J_A^\dagger\, \mathrm{d}J_A\, \rho_A\big] \le \|\mathrm{d}J_A\|_F^2, \qquad \rho_A = \sum_i |i\rangle\langle i| \otimes \gamma_{i,A},$$

where $\gamma_{i,A}$ is the reduced state of γ_i on Alice's factor; ρ_A is block diagonal with density matrices as blocks, so $\|\rho_A\|_{\mathrm{op}} \le 1$; in the scaled coordinates this is at most $\sqrt{S}$ times the coordinate speed, and the same holds for J_B. The tuple $(\gamma_i)_i$ contributes $\|\mathrm{d}C_\gamma\|_F = (\sum_i \|\dot\gamma_i\|^2)^{1/2}$, which is d times its coordinate speed. Since $r \le 2K/d^2$, $S \le 4K/d$ and $d^2 \le 2K/d \le K$, all these coefficients are at most $2\sqrt{K}$, so (38) holds. The rest of the proof of Lemma 6.2 applies verbatim, because the allowed directions for measurements have dimension at most $\ell_{\mathrm{m}} = 3d^2 - 2 \le 4d^2 - 3$. □

Theorem 6.7 (Haar volume near a maximally entangled basis)**.** *There is a universal C such that for $d \ge 2$, $K \ge d^3/4$ and $0 < \varepsilon \le 1/2$,*

$$\mu(\mathcal{M}_d^\dagger \cap \mathsf{Reach}^{\mathrm{m}}_{K,\varepsilon}) \le \min\{1,\ e^{CK^2}\, \varepsilon^{d^4/16}\}.$$

Proof. First let $\varepsilon \le 1/256$, so $\delta = \sqrt{50\varepsilon/9} \le 1$. If $K < d^3/2$ the set is empty by Lemma 6.5, so let $K \ge d^3/2$. We run the construction of Section 4.1 with the vectors γ_i of (40) in place of those of Lemma 3.2(ii), and with $\delta = \sqrt{50\varepsilon/9}$ in (22) and (24). The resulting witness families have $D = O(K^2)$ coordinates, bounded polynomial format, and at most e^{CK^2} members, and every target is within normalized distance δ of an image, by (24). Put $u = K/d^2 \ge d/2$ and apply Proposition 5.4 with the bounds of Lemma 6.6, as in the proof of Theorem 6.3. If $s \le c_0$, this gives

$$\mu(\mathcal{M}_d^\dagger \cap \mathsf{Reach}^{\mathrm{m}}_{K,\varepsilon}) \le e^{C(K^2+N)}\big[C\sqrt{1+u}\big]^N \delta^{N/8} \le e^{C'K^2}\varepsilon^{N/16},$$

because $N \le K^2$ and $N\ln(1+u) \le Nu = d^2K \le K^2$. If $s > c_0$, then $\ln(1/\varepsilon) \le C + \ln(1+u)$, and the claimed bound is at least one after enlarging C. For $1/256 < \varepsilon \le 1/2$ the bound is at least one once $C \ge 2$, since $K \ge d^3/4$ gives $N \le 4K^2$. □

Corollary 6.8 (Universal measurement approximation)**.** *For $d \ge 2$ and $0 < \varepsilon \le 1/2$, a footprint K that suffices for every rank-one PVM with $\delta_\Phi \le \varepsilon$ satisfies*

$$K \ge \max\left\{d^3(1-\varepsilon)^2,\ \frac{d^2\sqrt{\ln(1/\varepsilon)}}{36}\right\}.$$

For $d \ge 3$, the denominator 36 can be replaced by 27. For $d = 2^n$, this gives $\log_2 K \ge \max\{3n-2,\ 2n + \frac{1}{2}\log_2\ln(1/\varepsilon) - 6\}$, with 6 replaced by 5 when $n \ge 2$.

Proof. The floor for a maximally entangled basis (Lemma B.1) gives $K \ge d^3(1-\varepsilon)^2 \ge d^3/4$. Universality puts $\mathcal{M}_d^\dagger$ inside the reachable set, so (39) and Theorem 6.7 give $(1 + 8d^{3/2})^{-2N} \le e^{CK^2}\varepsilon^{N/16}$. Taking logarithms gives the precision term up to a universal constant: $K^2 \ge d^6/16$ absorbs the $N\ln d$ term from the patch mass. The numerical calculation in Appendix D gives 36 and 27. □

The measurement bound thus has the same precision term as the unitary bound, and a larger floor. The small radius t_d costs a factor $\ln d$ in the logarithm of the mass of the ball, which the floor $K \ge d^3/4$ absorbs.

6.3 Consequences and limits

Both regional estimates also give pointwise bounds for almost every target in their balls. Let $\varepsilon_j = e^{-j}$ and $K_j = \lfloor c\, d^2\sqrt{j} \rfloor$ with $c > 0$ small. For all large j, $K_j \geq d^2/2$ and $K_j \geq d^3/4$, so Theorems 6.3 and 6.7 bound the measures of $\mathcal{B}_d \cap \mathsf{Reach}^{\mathrm{u}}_{K_j,\varepsilon_j}$ and $\mathcal{M}_d^\dagger \cap \mathsf{Reach}^{\mathrm{m}}_{K_j,\varepsilon_j}$ by $e^{-(1/16 - Cc^2)d^4 j}$, which is summable. The Borel–Cantelli argument of Corollary 5.7 then shows that, for Haar-almost every $U \in \mathcal{B}_d$ and for the PVM of Haar-almost every basis matrix $M \in \mathcal{M}_d$, there is a threshold $\varepsilon_0 > 0$ below which $K_\varepsilon \geq c\, d^2 \sqrt{\ln(1/\varepsilon)}$, after decreasing c. The balls have positive but very small measure, so this does not improve Corollary 5.7, whose factor d comes from the whole-group estimate of Theorem 5.5. Neither regional estimate strengthens that whole-group estimate.

7 Free classical communication

The footprint charges classical messages, and this is what gives the strong dimension dependence of Theorems A and B. It is not needed for divergence with the precision. In this section classical communication is free: the classical messages may range over arbitrary, even continuous, alphabets, and we charge only the Schmidt number of the resource and the dimensions of the quantum messages.

Model. Every quantum register, including the quantum messages, has a fixed finite dimension. Party $X \in \{A, B\}$ first applies, to its input and its share of the resource, an instrument with outcomes in a standard Borel space Σ_X: a countably additive map $E \mapsto \mathcal{I}_X(E)$ from Borel sets $E \subseteq \Sigma_X$ to completely positive maps between fixed finite-dimensional matrix algebras, with $\mathcal{I}_X(\Sigma_X)$ trace preserving. Its quantum output consists of a kept register and an outgoing quantum message of dimension m^{q}_X. Both parties then send their quantum messages and broadcast their classical outcomes $x \in \Sigma_A$ and $y \in \Sigma_B$ simultaneously, and apply final channels that may depend on (x, y); we require their Choi matrices to be jointly Borel measurable in (x, y). We charge the *quantum footprint*

$$K_{\mathrm{q}} = r\, m^{\mathrm{q}}_A m^{\mathrm{q}}_B.$$

Local operations and simultaneous classical communication (LOSCC [GAVC25], also called LOBC [GC20]) is the case $m^{\mathrm{q}}_A = m^{\mathrm{q}}_B = 1$, where $K_{\mathrm{q}} = r$. Let $\mathsf{Reach}^{\mathrm{q,u}}_{K_{\mathrm{q}},\varepsilon}$ and $\mathsf{Reach}^{\mathrm{q,m}}_{K_{\mathrm{q}},\varepsilon}$ be the unitaries, respectively the matrices $M^\dagger$ of PVMs, that such protocols of quantum footprint at most K_{q} implement with $\delta_{\mathrm{Choi}} \leq \varepsilon$, respectively $\delta_\Phi \leq \varepsilon$; as before we drop the superscripts u and m when a statement applies to both. We write $K_{\mathrm{q},\varepsilon}(T)$ for the least quantum footprint of a target T at error ε. We use the Haar outer measure μ^*, since this class of protocols has no finite-dimensional parametrization. By Lemma 7.1 below, each of these reachable sets is contained in a compact reachable set $\mathsf{Reach}_{\overline{K},\varepsilon}$ of the footprint model, and the measure bounds below only use this containment.

Lemma 7.1 (Finite-support compression of the score)**.** *Consider a protocol in this model whose resource has Schmidt number at most r, and fix a target unitary or PVM with its score q_U or q_Φ. There is a protocol with*

(i) a pure resource of Schmidt rank $r' \leq r$;

(ii) the same quantum-message dimensions $m^{\mathrm{q}}_A, m^{\mathrm{q}}_B$;

(iii) finite classical alphabets $\Sigma'_X \subseteq \Sigma_X$ of sizes $|\Sigma'_X| \leq (dr')^2$;

(iv) a score at least that of the original protocol.

Copying the finite classical outcomes coherently into the messages turns it into a protocol of footprint at most

$$\overline{K} := d^4 K_{\mathrm{q}}^5. \tag{42}$$

Proof. The score is affine in the resource, so we select a pure component whose score is at least the average; let r' be its Schmidt rank and restrict the resource registers to the Schmidt supports. The input of party X's instrument then has dimension $s_X = dr'$.

We first compress Alice's outcomes. Let $\mathsf{J}_A(E) = \sum_{j,k} |j\rangle\langle k| \otimes \mathcal{I}_A(E)(|j\rangle\langle k|)$ be the unnormalized Choi matrix of the branch E. Then $\mu_A(E) = \operatorname{Tr} \mathsf{J}_A(E)/s_A$ is a probability measure, and finite-dimensional Radon–Nikodym gives a positive density $c_A(x)$ with $\mathsf{J}_A(E) = \int_E c_A \, d\mu_A$ and $\operatorname{Tr} c_A(x) = s_A$ almost everywhere. Let $\bar{c}_A(x)$ be the partial trace of $c_A(x)$ over the output; then $\operatorname{Tr} \bar{c}_A(x) = s_A$ and $\int \bar{c}_A \, d\mu_A = I$, which expresses trace preservation. Hold Bob's instrument and all final channels fixed, and let $f_A(x)$ be the contribution of Alice's branch x to the score, after averaging over Bob's outcome. It is measurable by Fubini's theorem, and $0 \le f_A(x) \le \|\bar{c}_A(x)\|_{\mathrm{op}} \le s_A$, since a contribution to the score is at most the probability of its branch. The pairs $(\bar{c}_A(x), f_A(x))$ lie in a real affine space of dimension $(s_A^2 - 1) + 1 = s_A^2$, and their μ_A-average is (I, q), where q is the score. By Carathéodory's theorem there are outcomes $x_1, \dots, x_{N_A}$ with $N_A \le s_A^2 + 1$ and probabilities p_j such that

$$\sum_j p_j \, \bar{c}_A(x_j) = I, \qquad \sum_j p_j \, f_A(x_j) = q.$$

The first identity says that the branches $p_j c_A(x_j)$, with the final channels of the outcomes x_j, form a trace-preserving instrument; the second says that the score is unchanged.

The selected values have to be attained at actual outcomes. Fix by Fubini a set Σ_A^0 of full μ_A-measure on which the density is defined and the required identities for Bob's outcomes hold, and make all selections in Σ_A^0. *Claim: the average (I, q) is a convex combination of at most $s_A^2 + 1$ values $(\bar{c}_A(x), f_A(x))$ with $x \in \Sigma_A^0$.* The average lies in the closed convex hull of these values. If it lies in the relative interior of that hull, finitely many attained values surround it, and Carathéodory reduces their number to at most $s_A^2 + 1$. If it lies on the relative boundary, a supporting hyperplane through it contains almost all values, and the claim follows by induction on the affine dimension.

If more than s_A^2 selected weights are positive, the corresponding Hermitian matrices are linearly dependent: choose real z_j, not all zero, with $\sum_j z_j \bar{c}_A(x_j) = 0$. Taking traces gives $\sum_j z_j = 0$; reverse the signs if necessary so that $\sum_j z_j f_A(x_j) \ge 0$. Replace p_j by $p_j + t z_j$, where $t = \min_{z_j < 0} p_j/(-z_j)$, restricting to the positive weights. The new weights are nonnegative, at least one vanishes, trace preservation is unchanged, and the score does not decrease. Thus at most s_A^2 outcomes suffice.

With Alice's finite instrument fixed, we perform the same selection and pruning for Bob inside the intersection of the finitely many full-measure sets associated with $x_1, \dots, x_{N_A}$. Neither step decreases the score, proving (iii)–(iv). This ordered choice is where joint measurability is used.

Each selected branch is a completely positive map with a finite Kraus representation, so its label can be copied coherently into one kept and one sent register. The complete outgoing messages then have dimensions $m_X^{\mathrm{q}} |\Sigma'_X|$. With $|\Sigma'_X| \le d^2 r'^2$ and $r' \le r' m_A^{\mathrm{q}} m_B^{\mathrm{q}} \le K_{\mathrm{q}}$, the footprint is

$$r' (m_A^{\mathrm{q}} |\Sigma'_A|)(m_B^{\mathrm{q}} |\Sigma'_B|) \le d^4 r'^4 \cdot r' m_A^{\mathrm{q}} m_B^{\mathrm{q}} \le d^4 K_{\mathrm{q}}^5. \qquad \square$$

The compression depends on the target and does not decrease its score; it need not preserve the channel. This suffices because every lower bound in these notes only uses the score (Remark 3.3).

The Schmidt-rank floors of Lemma B.1 survive free classical communication in a weaker form, because classical messages carry no Schmidt rank across the cut once they are conditioned on.

Lemma 7.2 (Floors with free classical communication)**.** *Consider a protocol with free classical messages and quantum footprint at most K_{q}.*

(i) For a unitary target U, $q_U \le \Lambda_{K_{\mathrm{q}}}(|U\rangle\!\rangle)$.

(ii) For a PVM with basis $(\varphi_i)_{i\le d^2}$, $q_\Phi \le \max_i \Lambda_{K_{\mathrm{q}}}(\varphi_i)$.

Consequently, approximating SWAP *with $\delta_{\mathrm{Choi}} \le \varepsilon$ requires $K_{\mathrm{q}} \ge d^2(1-\varepsilon)$, and approximating the measurement in a basis of maximally entangled vectors with $\delta_\Phi \le \varepsilon$ requires $K_{\mathrm{q}} \ge d(1-\varepsilon)$. Almost every unitary U requires $K_{\mathrm{q}} \ge d^2$ when $\varepsilon < \varpi_{\min}(U)$, and almost every PVM requires $K_{\mathrm{q}} \ge d$ when $\varepsilon < \lambda_*$, with $\varpi_{\min}$ and λ_* as in Lemma B.1(iii).*

Proof. By Lemma 7.1, which does not decrease the score and keeps the quantum-message dimensions, we may assume a pure resource of Schmidt rank at most r and finite classical alphabets. Run the protocol on the Choi input and fix a pair of classical outcomes and Kraus operators of all local operations, including the discarding of garbage. The resulting unnormalized vector on $A'\bar{A}$ and $B'\bar{B}$ has Schmidt rank at most $r\, m_A^{\mathrm{q}} m_B^{\mathrm{q}} \le K_{\mathrm{q}}$ across the laboratory cut: as in Lemma 3.1, only the quantum messages cross the cut, and the classical outcomes are fixed. The Choi state ρ of the implemented channel is the finite sum of the rank-one operators of these vectors, so it has Schmidt number at most K_{q}. (i) Hence $q_U = \langle\!\langle U|\rho|U\rangle\!\rangle \le \Lambda_{K_{\mathrm{q}}}(|U\rangle\!\rangle)$ by (11). (ii) Let $\sigma_i = (\langle i|_{A'}\langle i|_{B'} \otimes I)\rho(|i\rangle_{A'}|i\rangle_{B'} \otimes I)$ on $\bar{A}\bar{B}$. Projecting onto a product vector is a local operation, so each σ_i has Schmidt number at most K_{q}, and $\sum_i \operatorname{Tr}\sigma_i \le 1$. Since $|\Omega\rangle = d^{-1}\sum_i |\varphi_i\rangle|\bar{\varphi}_i\rangle$,

$$q_\Phi = \sum_i \langle\bar{\varphi}_i|\sigma_i|\bar{\varphi}_i\rangle \le \sum_i \operatorname{Tr}(\sigma_i)\,\Lambda_{K_{\mathrm{q}}}(\varphi_i) \le \max_i \Lambda_{K_{\mathrm{q}}}(\varphi_i).$$

The consequences follow as in Lemma B.1: $\Lambda_k(|\mathsf{SWAP}\rangle\!\rangle) = \min\{k/d^2, 1\}$, a maximally entangled vector has $\Lambda_k = \min\{k/d, 1\}$, and if K_{q} is below the Schmidt rank then $\Lambda_{K_{\mathrm{q}}}$ is at most one minus the smallest squared Schmidt coefficient. □

With classical messages free, the labels no longer need to be charged, which is why the floor for measurements drops from d^3 to d.

Corollary 7.3 (Free classical communication)**.** *There are universal $c, C > 0$ such that, for $d \ge 2$, $K_{\mathrm{q}} \ge 1$ and $0 < \varepsilon \le 1/2$,*

$$\mu^*(\mathsf{Reach}^{\mathrm{q,u}}_{K_{\mathrm{q}},\varepsilon}) \le \min\{1,\ e^{Cd^{10}K_{\mathrm{q}}^{10}\varepsilon^{\kappa_{\mathrm{u}}/2}}\}, \qquad \mu^*(\mathsf{Reach}^{\mathrm{q,m}}_{K_{\mathrm{q}},\varepsilon}) \le \min\{1,\ e^{Cd^{10}K_{\mathrm{q}}^{10}\varepsilon^{\kappa_{\mathrm{m}}/2}}\}. \quad (43)$$

Consequently:

(i) a quantum footprint that suffices for every unitary, respectively for every rank-one PVM, satisfies

$$K_{\mathrm{q}} \ge \max\{d^2(1-\varepsilon),\ c\,d^{-2/5}[\ln(1/\varepsilon)]^{1/10}\} \quad \textit{(unitaries)},$$
$$K_{\mathrm{q}} \ge \max\{d(1-\varepsilon),\ c\,d^{-2/5}[\ln(1/\varepsilon)]^{1/10}\} \quad \textit{(PVMs)};$$

for $d = 2^n$, in both cases $\log_2 K_{\mathrm{q}} \ge \frac{1}{10}\log_2\ln(1/\varepsilon) - \frac{2}{5}n - O(1)$;

(ii) for each $d \ge 2$, almost every unitary and almost every rank-one PVM T has a threshold $\varepsilon_0(T) > 0$ such that $K_{\mathrm{q},\varepsilon}(T) \ge c\,d^{-3/5}[\ln(1/\varepsilon)]^{1/10}$ for $0 < \varepsilon < \varepsilon_0(T)$.

In LOSCC, $K_{\mathrm{q}} = r$, and a resource of Schmidt number r occupies at least $2\log_2 r$ qubits.

Proof. By Lemma 7.1, $\mathsf{Reach}^{\mathrm{q}}_{K_\mathrm{q},\varepsilon}$ is contained in the compact set $\mathsf{Reach}_{\overline{K},\varepsilon}$, where $\overline{K} = d^4K_\mathrm{q}^5 \geq d^2$. Theorem 5.5 with $d^2\overline{K}^2 = d^{10}K_\mathrm{q}^{10}$ gives (43). For (i), the floors are those of Lemma 7.2; for the precision terms, universality in this model gives universality in the footprint model at footprint $\overline{K}$, and Corollaries 6.4 and 6.8 give $c\,d^2\sqrt{\ln(1/\varepsilon)} \leq \overline{K} = d^4K_\mathrm{q}^5$. For (ii), let $\varepsilon_j = e^{-j}$ and $K_j = \lfloor c_1 d^{-3/5} j^{1/10} \rfloor$ with c_1 small; for large j, $K_j \geq 1$ and $d^{10}K_j^{10} \leq c_1^{10}d^4 j$, so the bounds (43) are at most $e^{-c_2 d^4 j}$. Borel–Cantelli for outer measures and monotonicity in K_q and ε conclude as in Corollary 5.7. □

Localizable measurements. In an *approximate localization* of a PVM Φ, Alice and Bob share a finite-dimensional state of Schmidt number r, do not communicate, and measure their inputs together with their shares of the state, with outcomes x and y in standard Borel spaces, for instance finite sets. A fixed measurable function $f(x,y) \in \{1,\dots,d^2\}$ is the reported outcome, and the failure probability is $1 - d^{-2}\sum_i \Pr[f(x,y) = i \mid \varphi_i]$. In [FGMS26] we showed that a PVM in a Haar-random basis almost surely has no exact localization with a finite-dimensional resource and finite outcome sets; the following is the quantitative version.

Corollary 7.4 (Approximate localization)**.** *With the universal constants c, C of Corollary 7.3, the following holds for each $d \geq 2$. Almost every rank-one PVM on $\mathbb{C}^d \otimes \mathbb{C}^d$ has a threshold $\varepsilon_0 > 0$ such that every localization with failure probability $\varepsilon < \varepsilon_0$ uses a resource of Schmidt number*

$$r \geq \max\{d,\ c\,d^{-3/5}[\ln(1/\varepsilon)]^{1/10}\}.$$

Moreover, for $0 < \varepsilon \leq 1/2$, the PVMs that have a localization with Schmidt number at most r and failure probability at most ε have outer measure at most $\min\{1, e^{Cd^{10}r^{10}}\varepsilon^{\kappa_\mathrm{m}/2}\}$.

Proof. As in [FGMS26], the parties broadcast x and y and both output $f(x,y)$. This is a protocol with free classical messages and no quantum messages, so $K_\mathrm{q} = r$, and its two outputs always agree, so its score q_Φ equals the success probability of the localization. Apply Lemma 7.2 and Corollary 7.3. □

The entanglement cost and the structure of exactly localizable measurements are studied in [PPKDG25, PBPKG26, AM26]; Corollary 7.4 is a precision-dependent bound for approximate localization of a generic basis.

8 Application: diagonal gates at fixed dimension

At fixed local dimension, Corollary 5.7 says that $\log_2 K$ must grow at least like $\log\log(1/\varepsilon)$ for almost every target, and Corollary 7.3 says the same for the quantum footprint, and hence for the number of resource qubits when only classical messages are sent. Here we show that the same lower bounds hold within the family of diagonal gates, which has Haar measure zero and is therefore not covered by the whole-group estimates.

Model and cost. In Theorem 8.1 the error is the normalized diamond distance, except for two qubits, where the Choi infidelity suffices; the reason is the restriction step in its proof. We use three cost measures: the footprint K of Section 2.3; the quantum footprint K_q of Section 7, when classical messages are free; and, when there are only classical messages (LOSCC), the number of resource qubits. Let $n_\varepsilon^{\mathrm{res}}(U)$ be the least total number of initial resource qubits of an LOSCC protocol that implements U with normalized diamond error at most ε, where shared randomness and fresh local workspace are free and the number of qubits is bounded in every branch of the shared randomness. A pure resource of Schmidt rank r occupies at least $2\log_2 r$ qubits.

Theorem 8.1 (Diagonal gates)**.** *Fix $d_A, d_B \geq 2$. For almost every diagonal unitary $U|a,b\rangle = e^{i\theta_{ab}}|a,b\rangle$ on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$, with respect to the uniform measure on the phases (θ_{ab}), there is $\varepsilon_0(U) > 0$ such that for $0 < \varepsilon < \varepsilon_0(U)$:*

(i) every protocol of footprint K that implements U with normalized diamond error at most ε satisfies $\log_2 K \geq \frac{1}{2}\log_2 \ln(1/\varepsilon) - O(1)$;

(ii) every protocol with free classical messages and quantum footprint K_{q} that implements U with normalized diamond error at most ε satisfies $\log_2 K_{\mathrm{q}} \geq \frac{1}{10}\log_2 \ln(1/\varepsilon) - O(1)$; hence $n_\varepsilon^{\mathrm{res}}(U) \geq \frac{1}{5}\log_2 \ln(1/\varepsilon) - O(1)$.

The $O(1)$ terms are universal. For $d_A = d_B = 2$ the same holds with the Choi infidelity in place of the diamond error, in particular for the controlled phase C_θ for almost every θ. For all sufficiently small ε, the same bounds hold for the worst case over all diagonal unitaries.

8.1 One scalar invariant

We follow the operator purity of [FGMS26], now evaluated on arbitrary, not necessarily unitary, matrices. For a complex $d^2 \times d^2$ matrix X let

$$P(X) = \frac{1}{d^4}\,\mathrm{Tr}\big[(X \otimes X)\, S_A\, (X \otimes X)^\dagger\, S_A\big], \tag{44}$$

where S_A swaps the two copies of H_A in $(H_A \otimes H_B)^{\otimes 2}$. The proof of the operator-purity lemma of [FGMS26] applies verbatim to arbitrary matrices and shows that $P((L_A \otimes L_B)X(R_A \otimes R_B)) = P(X)$ for all local unitaries. Differentiating along $t \mapsto e^{-tb}Xe^{ta}$ with $a, b \in \mathfrak{u}_{\mathrm{loc}}$ gives

$$\mathrm{d}P_X[-bX + Xa] = 0 \qquad \text{for all matrices } X \text{ and all } a, b \in \mathfrak{u}_{\mathrm{loc}}. \tag{45}$$

For the two-qubit controlled phase $C_\theta = \mathrm{diag}(1,1,1,e^{i\theta})$, which is the case $n = 1$ of the controlled phase of [FGMS26], the same lemma gives $P(C_\theta) = (3 + \cos\theta)/4$. Since P is a sum of four-fold products of X and $X^\dagger$ and $|\mathrm{Tr}(YS_AZS_A)| \leq \|Y\|_F\|Z\|_F$,

$$|\mathrm{d}P_X[V]| \leq \frac{4}{d^4}\|X\|_F^3\|V\|_F, \tag{46}$$

which for $d = 2$ and $\|X\|_F \leq 2$ is at most $2\|V\|_F$.

Proposition 8.2 (Controlled-phase length bound)**.** *Let J be a compact interval in $(0,\pi)$ or in $(\pi, 2\pi)$. The set of $\theta \in J$ such that C_θ has a protocol of footprint at most K with $\delta_{\mathrm{Choi}} \leq \varepsilon$ has Lebesgue measure at most $C_J e^{CK^2}\sqrt{\varepsilon}$. The set of $\theta \in J$ such that C_θ has a protocol with free classical messages, quantum footprint at most K_{q} and Choi infidelity at most ε has outer measure at most $C_J e^{CK_{\mathrm{q}}^{10}}\sqrt{\varepsilon}$.*

Proof. Here $d = 2$, so the witness families of Proposition 5.1 have $D = O(K^2)$ coordinates. Every approximable C_θ has a witness $x \in \widetilde{\mathcal{Z}}_\delta$ with $\|\widehat{T}(x) - C_\theta\|_{\mathrm{n}} \leq \delta \leq 2\sqrt{\varepsilon}$, and $\|\widehat{T}\|_F \leq \|\mathsf{Dec}\|_{\mathrm{op}}\|\mathsf{Enc}\|_F = 2$. By Proposition 4.2 and (45), the allowed part of the velocity does not change P, so

$$|\mathrm{d}(P \circ \widehat{T})_x(\dot{x})| = |\mathrm{d}P_{\widehat{T}}[\Xi(\dot{x})]| \leq 2\|\Xi(\dot{x})\|_F \leq 2\sqrt{h}\,\beta\|\dot{x}\| \leq C\delta D\|\dot{x}\|,$$

using (46), (29) and $h \leq 4\delta^2$. By (46) on the segment from $\widehat{T}(x)$ to C_θ, also $|P(\widehat{T}(x)) - P(C_\theta)| \leq 2\|\widehat{T}(x) - C_\theta\|_F \leq 4\delta$. Consider $g(x,t) = P(\widehat{T}(x)) + 4\delta t$ on $\widetilde{\mathcal{Z}}_\delta \times [-1,1]$. Every value $P(C_\theta)$ of an approximable phase lies in its image. Using the polynomial extension of $\widehat{T}$, the map g has degree at most 72, its source has bounded polynomial format in $D+1$ variables and universal radius, and its ambient derivative on that source has norm at most $C\delta D$. Lemma 5.3

with $m = 1$, summed over the e^{CD} families, bounds the length of the set of these values by $e^{CD}\delta \le e^{CK^2}\sqrt{\varepsilon}$. On J the map $\theta \mapsto P(C_\theta) = (3 + \cos\theta)/4$ is monotone with derivative of absolute value at least $\min_J |\sin\theta|/4 > 0$, which proves the first claim.

For the second claim, Lemma 7.1 turns such a protocol into a protocol of footprint at most $2^4 K_{\mathrm{q}}^5 = 16 K_{\mathrm{q}}^5$ whose score is at least as large. Substituting $K = 16K_{\mathrm{q}}^5$ proves the bound, since the score does not decrease. □

Proof of Theorem 8.1. Controlled phases. Let $\varepsilon_j = e^{-j}$, $K_j = \lfloor c\sqrt{j} \rfloor$ and $K_{\mathrm{q},j} = \lfloor c\, j^{1/10} \rfloor$ with a universal $c > 0$ small enough that both bounds of Proposition 8.2 are at most $C_J e^{-j/4}$. These bounds are summable, so by the Borel–Cantelli lemma, for outer measure in the second case, almost every $\theta \in J$ admits, for all large j, no protocol with Choi infidelity ε_j and footprint at most K_j, and no protocol with free classical messages, Choi infidelity ε_j and quantum footprint at most $K_{\mathrm{q},j}$. Monotonicity in ε and a countable exhaustion of $(0,\pi) \cup (\pi, 2\pi)$ by intervals J give both bounds for C_θ, with the Choi infidelity, for almost every θ, and with universal constants. In LOSCC, $K_{\mathrm{q}} = r$, and a protocol with shared randomness is a mixture of pure-resource protocols with at most q qubits each, so its resource has Schmidt number at most $2^{q/2}$; this gives the bound on $n_\varepsilon^{\mathrm{res}}$. The diamond error bounds the Choi infidelity by (9).

Two qubits. Every diagonal two-qubit unitary equals, up to a global phase, $(D_A \otimes D_B)\, C_\theta$ with diagonal unitaries D_A, D_B and

$$\theta = \theta_{11} - \theta_{10} - \theta_{01} + \theta_{00} \pmod{2\pi}.$$

Composing the decoders with $D_A^\dagger$ and $D_B^\dagger$ changes neither the score nor the resources, and the map from the phases to θ sends the uniform measure to the uniform measure on the circle. This proves the theorem for $d_A = d_B = 2$, with the Choi infidelity.

General dimensions. Let Π_X be the projection of $\mathbb{C}^{d_X}$ onto the span of $|0\rangle$ and $|1\rangle$. Given a protocol for U, feed it only inputs in the range of $\Pi_A \otimes \Pi_B$, a local isometric embedding of $\mathbb{C}^2 \otimes \mathbb{C}^2$, and apply to each output the channel $\rho \mapsto \Pi_X \rho \Pi_X + \mathrm{Tr}[(I - \Pi_X)\rho]\, \sigma_X$ for a fixed state σ_X on the two levels. Since U maps the range of $\Pi_A \otimes \Pi_B$ to itself, this gives a protocol for the two-qubit gate $\mathrm{diag}(e^{i\theta_{00}}, e^{i\theta_{01}}, e^{i\theta_{10}}, e^{i\theta_{11}})$. Composing with channels does not increase the normalized diamond distance, and local pre- and post-processing does not change the resources. The Choi infidelity can increase under this restriction, which is why we use the diamond error in this section. Since the four phases $\theta_{00}, \dots, \theta_{11}$ are uniformly distributed, the two-qubit case gives the theorem for almost every U.

Worst case. Both bounds of Proposition 8.2 are smaller than the length of J once $K \le c\sqrt{\ln(1/\varepsilon)}$, respectively $K_{\mathrm{q}} \le c[\ln(1/\varepsilon)]^{1/10}$, and ε is small enough. Then some C_θ with $\theta \in J$ has no protocol with these resources and Choi infidelity at most ε. By the two steps above, no diagonal unitary whose restriction to the levels 0 and 1 equals C_θ up to local diagonal unitaries and a global phase has a protocol with these resources and normalized diamond error at most ε. □

Remark 8.3 (Exceptional gates). The lower bound is not pointwise. Gates with exact finite protocols have bounded cost at every precision, and they are dense in the controlled-phase circle: C_θ has an exact protocol whenever $\theta \in \pi\mathbb{Q}$ [Yu11, YGC12]. For the named gate C_1, which has no exact protocol, Theorem C.1 in Appendix C gives only a triple-logarithmic lower bound on the number of resource qubits.

Remark 8.4 (Two-qubit gates). For general two-qubit gates the whole-group bounds apply directly: for Haar-almost every two-qubit unitary and all ε below a threshold, $\log_2 K_\varepsilon(U) \ge \frac{1}{2}\log_2 \ln(1/\varepsilon) - O(1)$ by Corollary 5.7, and $n_\varepsilon^{\mathrm{res}}(U) \ge \frac{1}{5}\log_2 \ln(1/\varepsilon) - O(1)$ by Corollary 7.3(ii), since $n^{\mathrm{res}} \ge 2\log_2 r$ and $K_{\mathrm{q}} = r$ in LOSCC.

9 Formal verification

The formalization [dMU21, mC20, NLQ26], using Lean and Mathlib 4.35.0, proves Theorems A to C, the diagonal-gate bounds, fixed-budget stability, qualitative score gaps and Schmidt-rank floors, including the numerical universal bounds of Appendix D. The effective bounds of Theorem C.1 for C_1 are also proved without any unproved mathematical input. The repository coverage guide lists declarations, operational-error variants and the precise correspondence with the manuscript.

Arithmetic bounds. All conclusions of Theorem C.1 for C_1 are proved, including the $R^4K_q^2$ trade-off, sixth-power free-classical bound and logarithmic and LOSCC consequences. For other nonzero real algebraic angles, the formal results give the charged bound and a tenth-power free-classical bound; the sixth-power extension in the final remark of Appendix C is not yet formalized. The proof constructs a polynomial relation for the optimal score directly, with single-exponential degree and bit-size bounds, instead of invoking effective quantifier elimination; Remark C.2 outlines the construction. It combines this with a proved Gelfond transcendence measure, with exponent 50 for general nonzero real algebraic angles and 33 for C_1. For C_1, the formal charged bound gives the explicit estimate

$$\epsilon \geq \exp[-\exp(9400K^2)] \qquad (K \geq 1)$$

for every protocol of footprint at most K with $\delta_{\mathrm{Choi}} \leq \epsilon$. With free classical messages, it gives $\epsilon \geq \exp[-\exp(2406400K_q^{10})]$. The sharper $R^4K_q^2$ trade-off, sixth-power and LOSCC bounds for C_1, and the bounds for other algebraic angles, are formalized with existential constants; their effective computability is not formalized. Kernel audits use only Lean's standard logical axioms: `propext`, `Classical.choice` and `Quot.sound`.

Scope. The charged results allow arbitrary finite private registers, pure resources and common-map finite mixtures. They use the scores of Section 2.2, with normalized diamond and joint total-variation variants; Haar estimates use outer measure where measurability is not formalized. The PVM targets are basis matrices M rather than $M^\dagger$, an equivalent convention by inversion invariance of Haar measure. Fixed-budget stability includes both unitary and PVM targets and their finite-orbit decompositions.

Free-classical coverage includes standard-Borel outcomes, the compression to $(dr')^2$ outcomes per party and charged footprint $d^4K_q^5$, localization, and measurable shared randomness with branch-dependent finite systems and almost-everywhere footprint caps. This includes the C_1 sixth-power bound and the diagonal-gate results; for diagonal gates the formal worst-case statement gives one hard gate per dimension. Almost-everywhere error thresholds remain target-dependent.

Proof differences. The image-volume estimate is proved for every polynomial format, and the semialgebraic Łojasiewicz inequality is also proved. The barrier argument supplies the constants used in Appendix D; the library additionally contains the active-constraint proof of Appendix A. The compression distances agree with the manuscript. The near-Bell bound uses the transverse derivative estimate directly, although Lemma 6.6 is separately formalized. Singular-value bounds are expressed equivalently as a sum of a bounded-rank map and a small-norm map. Reachable-space compression preserves the implemented channel rather than the chosen dilation, and attainment is proved for optimal scores, not minimal operational errors.

The sharper Cijsouw measure (57) [Cij74, Theorem 1] and the resulting certificate estimate (59) are not formalized in their written form. For C_1 the formal substitute is

$$g_* \geq \exp[-202\,(6L + \tau + 128)^{33}],$$

where L and τ are the boundary polynomial's degree and bit size. It gives the same final double-exponential rate. The general written image-volume estimate likewise has intermediate constants not individually verified; the numerical bounds reported in Appendix D have checked counterparts.

10 Discussion and open questions

The exact results of [FGMS26] and the bounds of these notes come from one calculation. A frozen environment makes the compressed generators local, or diagonal for measurements, and the derivative of the overlap between the encoder and decoder isometries differs from these allowed motions only by a leakage term. At zero leakage, integration fixes a local-unitary orbit; at nonzero leakage, a bounded number of coordinates and a dimension-uniform tube estimate limit how much of the target space can be approximated. We close with some questions that this leaves open.

Fixed error and growing dimension. Our bounds concern the dependence on the precision. At constant error they only give the floors $K = \Omega(d^2)$ for universal approximation of unitaries and $K = \Omega(d^3)$ for measurements, whereas the known protocols for general unitaries use resources exponential in d. Proving super-polynomial lower bounds at fixed error remains the central open problem for NLQC and position verification [May26]; the geometric approach here would need a much finer control of the image of the protocol map than the counting of its parameters.

The dimension dependence of the precision term. Theorem A compares $O(d^2K^2)$ parameters with $\Theta(d^4)$ transverse directions, which gives the factor d; near SWAP and near a maximally entangled basis, Theorem B improves this to d^2 for universal approximation. We do not know whether $d^2\sqrt{\ln(1/\varepsilon)}$ is the correct universal order, whether the almost-everywhere bounds can also be improved to d^2, or whether the power $\sqrt{\ln(1/\varepsilon)}$ is optimal for general targets. A Hilbert–Schmidt version of the singular-value cutoff in Section 6 appears to restore the full exponent $\kappa_{\mathrm{u}}/2$ in the estimate near SWAP; this would not change the universal bounds.

Free classical communication. The geometric bounds with free classical communication use the transcript compression of Lemma 7.1, losing a fifth power of the footprint and introducing the factors $d^{-2/5}$ and $d^{-3/5}$. The loss comes from charging the compressed transcript as a quantum message and then treating the result as an arbitrary protocol. A witness that keeps the transcript as orthogonal classical blocks, with one frozen environment and one completed decoder per transcript, appears to need only $O(d^6K_{\mathrm{q}}^6)$ coordinates, which would improve the exponent $\frac{1}{10}$ to $\frac{1}{6}$; we have not written this out. The direct physical parametrization in Appendix C already gives a sixth-power arithmetic bound for C_1; the generic improvement additionally requires the derivative and volume estimates. Removing the loss entirely would bring the bounds with free classical communication close to those with charged messages.

Entropy. All our bounds concern Schmidt numbers and message dimensions, and they give no lower bound on the entanglement entropy of the resource. It remains open whether the targets excluded in [FGMS26] can be implemented exactly with a resource of infinite Schmidt rank but finite entanglement entropy, together with finite-dimensional messages.

Explicit targets. For C_1, Appendix C gives $\log_2 K = \Omega(\log\log\log(1/\varepsilon))$, while almost every controlled phase requires $\log_2 K = \Omega(\log\log(1/\varepsilon))$ (Theorem 8.1). In the arithmetic argument, the remaining loss is the exponential degree and logarithmic height of the boundary

polynomial: polynomial bounds on both would give $\varepsilon \geq \exp[-\operatorname{poly}(K)]$ by (59), giving $\log_2 K = \Omega(\log\log(1/\varepsilon))$. Such control, or a different method for named targets, is still needed.

Other models. The whole-group results of Sections 5 to 7 are stated for balanced local dimensions; the absorption of dimension factors in the proof of Theorem 5.5 is not uniform in the aspect ratio d_A/d_B, and unbalanced dimensions need a separate argument. The exact argument is expected to extend to any fixed number of parties [FGMS26]; a quantitative multipartite theorem would in addition need a suitable resource measure and a joint compression of the environment. For general channels, and for rank-one POVMs, whose analysis operators have additional directions of motion, the frozen dilation is weaker and the argument has to be modified.

Position verification. Our volume bound, Theorem A(i), averages over targets, while security of position verification requires lower bounds for the verifier's actual task and success criterion, typically at nonzero error on a fixed input distribution. The scores we use are of this form for the uniform input distribution on a basis or on half of a maximally entangled state, but the targets of position-verification protocols are specific and are not covered by the almost-everywhere statements.

AI statement

These notes are currently an almost-entirely AI-written exposition, reporting explorations of possible extensions to the results of [FGMS26]. Various models, including OpenAI GPT-6 Astra, Anthropic Claude Fable 5.1, and Opus 5.5 were involved in proving the results and creating these notes, in August and September 2026, following the authors' direction and selection.

References

[ACCM25] Vahid Asadi, Richard Cleve, Eric Culf, and Alex May. Linear gate bounds against natural functions for position-verification. *Quantum*, 9:1604, 2025. Corrected version: arXiv:2402.18648v5 (2026).

[ACM25] Vahid R. Asadi, Eric Culf, and Alex May. Rank lower bounds on non-local quantum computation. In Raghu Meka, editor, *16th Innovations in Theoretical Computer Science Conference (ITCS 2025)*, volume 325 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 11:1–11:18. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025.

[AM26] Seiseki Akibue and Jisho Miyazaki. Localization of joint quantum measurements on $\mathbb{C}^d \otimes \mathbb{C}^d$ by entangled resources with Schmidt number at most d. *Physical Review A*, 113(5):052430, 2026.

[BCF+14] Harry Buhrman, Nishanth Chandran, Serge Fehr, Ran Gelles, Vipul Goyal, Rafail Ostrovsky, and Christian Schaffner. Position-based quantum cryptography: Impossibility and constructions. *SIAM Journal on Computing*, 43(1):150–178, 2014.

[BCR98] Jacek Bochnak, Michel Coste, and Marie-Françoise Roy. *Real Algebraic Geometry*, volume 36 of *Ergebnisse der Mathematik und ihrer Grenzgebiete (3)*. Springer-Verlag, Berlin, 1998.

[BCS22] Andreas Bluhm, Matthias Christandl, and Florian Speelman. A single-qubit position verification protocol that is secure against multi-qubit attacks. *Nature Physics*, 18(6):623–626, 2022.

[Bha97] Rajendra Bhatia. *Matrix Analysis*, volume 169 of *Graduate Texts in Mathematics*. Springer, New York, 1997.

[BK11] Salman Beigi and Robert König. Simplified instantaneous non-local quantum computation with applications to position-based cryptography. *New Journal of Physics*, 13(9):093036, 2011.

[Bog26] Kevin Bogner. Robust logarithmic entanglement lower bound for f-routing. arXiv:2608.05775, 2026.

[BPR06] Saugata Basu, Richard Pollack, and Marie-Françoise Roy. *Algorithms in Real Algebraic Geometry*. Springer, Berlin, 2nd edition, 2006.

[Cij74] P. L. Cijsouw. Transcendence measures of exponentials and logarithms of algebraic numbers. *Compositio Mathematica*, 28(2):163–178, 1974.

[CLM+21] Matthias Christandl, Felix Leditzky, Christian Majenz, Graeme Smith, Florian Speelman, and Michael Walter. Asymptotic performance of port-based teleportation. *Communications in Mathematical Physics*, 381(1):379–451, 2021.

[CM26] Richard Cleve and Alex May. Lower bounds on non-local computation from controllable correlation. arXiv:2602.00255, 2026.

[dMU21] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, volume 12699 of *Lecture Notes in Computer Science*, pages 625–635. Springer, 2021.

[Fed69] Herbert Federer. *Geometric Measure Theory*, volume 153 of *Grundlehren der mathematischen Wissenschaften*. Springer, New York, 1969.

[FGMS26] Marten Folkertsma, Dmitry Grinko, Gina Muuss, and Florian Speelman. Perfect non-local quantum computation is impossible. arXiv:2609.40228, 2026.

[FNA+20] Philippe Faist, Sepehr Nezami, Victor V. Albert, Grant Salton, Fernando Pastawski, Patrick Hayden, and John Preskill. Continuous symmetries and approximate quantum error correction. *Physical Review X*, 10(4):041018, 2020.

[GAVC25] Ian George, Rene Allerstorfer, Philip Verduyn Lunel, and Eric Chitambar. Orthogonality broadcasting and quantum position verification. *New Journal of Physics*, 27:054511, 2025.

[GC20] Alvin Gonzales and Eric Chitambar. Bounds on instantaneous nonlocal quantum computation. *IEEE Transactions on Information Theory*, 66(5):2951–2963, 2020.

[Haf23] Jonas Haferkamp. On the moments of random quantum circuits and robust quantum complexity. arXiv:2303.16944, 2023.

[HFK+22] Jonas Haferkamp, Philippe Faist, Naga B. T. Kothakonda, Jens Eisert, and Nicole Yunger Halpern. Linear growth of quantum circuit complexity. *Nature Physics*, 18:528–532, 2022.

[JKPPG22] M. Junge, A. M. Kubicki, C. Palazuelos, and D. Pérez-García. Geometry of Banach spaces: a new route towards position based cryptography. *Communications in Mathematical Physics*, 394:625–678, 2022.

[LRT25] Antonio Lerario, Luca Rizzi, and Daniele Tiberio. Quantitative approximate definable choices. *Mathematische Annalen*, 392(1):1289–1319, 2025. arXiv:2409.14869; theorem numbers as in arXiv v2.

[May26] Alex May. Entanglement cost in non-local quantum computation. arXiv:2605.02840, 2026.

[mC20] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 367–381. ACM, 2020.

[MCPPG26] I. M. Moreno-Cuadrado, C. Palazuelos, and D. Pérez-García. Exponential lower bounds for low-degree strategies in position-based quantum cryptography. arXiv:2609.35592, 2026.

[NLQ26] NLQC-lean: Lean proofs of resource bounds for one-round non-local quantum computation. https://github.com/fspeelman/NLQC, 2026. Lean 4 library, 3 October 2026.

[PBPKG26] Jef Pauwels, Cyril Branciard, Alejandro Pozas-Kerstjens, and Nicolas Gisin. Symmetric localizable multipartite quantum measurements from Pauli orbits. *Physical Review A*, 113(5):052432, 2026.

[PPKDG25] Jef Pauwels, Alejandro Pozas-Kerstjens, Flavio Del Santo, and Nicolas Gisin. Classification of joint quantum measurements based on entanglement cost of localization. *Physical Review X*, 15(2):021013, 2025.

[SG11] Dan Stahlke and Robert B. Griffiths. Entanglement requirements for implementing bipartite unitary operations. *Physical Review A*, 84:032316, 2011.

[TFKW13] Marco Tomamichel, Serge Fehr, Jędrzej Kaniewski, and Stephanie Wehner. A monogamy-of-entanglement game with applications to device-independent quantum cryptography. *New Journal of Physics*, 15(10):103002, 2013.

[Vai03] Lev Vaidman. Instantaneous measurement of nonlocal variables. *Physical Review Letters*, 90(1):010402, 2003.

[WCPG11] Michael M. Wolf, Toby S. Cubitt, and David Pérez-García. Are problems in quantum information theory (un)decidable? arXiv:1111.5425, 2011.

[YC04] Yosef Yomdin and Georges Comte. *Tame Geometry with Application in Smooth Analysis*, volume 1834 of *Lecture Notes in Mathematics*. Springer, Berlin, 2004.

[YGC12] Li Yu, Robert B. Griffiths, and Scott M. Cohen. Fast protocols for local implementation of bipartite nonlocal unitaries. *Physical Review A*, 85(1):012304, 2012.

[Yu11] Li Yu. Fast controlled unitary protocols using group or quasigroup structures. arXiv:1112.0307, 2011. Version 3, revised 2013.

A Proof of the tube estimate

In this appendix all matrix norms are ordinary Frobenius norms, so $\mathbb{C}^{d^2\times d^2} \cong \mathbb{R}^{2N}$ with $N = d^4$. We write ω_m for the volume of the unit ball in $\mathbb{R}^m$ and $U_\tau(E)$ for the closed Frobenius τ-neighbourhood of a set E. The argument has three parts: a Jacobian bound for polynomial images (Lemma 5.3), a comparison of ambient volume with Haar measure (Lemma A.2), and the application to a thickened image.

A.1 Proof of the image-volume lemma

We first record an elementary bound for isolated polynomial zeros. It is enough for the argument below; no bound on the number of connected components of a general semialgebraic set is needed.

Lemma A.1 (Isolated zeros)**.** *Let F be a nonnegative real polynomial of degree at most $2q$ in n variables, where $n, q \geq 1$ are integers. Then F has at most $(2q+1)^n$ isolated real zeros.*

Proof. Put $e = 2q+1$ and first consider a polynomial system

$$a_i x_i^e + h_i(x) = 0 \quad (1 \leq i \leq n), \qquad a_i \neq 0, \quad \deg h_i < e.$$

On its common zero set, replacing x_i^e by $-h_i/a_i$ lowers total degree. Induction on total degree therefore expresses the restriction of every polynomial as a linear combination of the e^n monomials x^α with $0 \leq \alpha_i < e$. For any finite set Z of distinct common zeros, polynomial evaluation onto $\mathbb{R}^Z$ is surjective: for each $z \in Z$, take a product of affine linear factors that equals one at z and vanishes at every other point of Z. Thus $|Z| \leq e^n$. Applying this to every finite subset bounds the entire common zero set by e^n.

Now take any finite set of isolated zeros of F, and put disjoint closed balls around them, small enough that F is positive on each boundary. For all sufficiently small $\varepsilon > 0$, the polynomial

$$F_\varepsilon(x) = F(x) + \varepsilon \sum_{i=1}^n x_i^{2q+2}$$

has a minimum in the interior of each ball: its value at the central zero tends to zero, whereas its boundary values are bounded below by a positive constant. The resulting minima are distinct common zeros of

$$\varepsilon(2q+2)x_i^{2q+1} + \partial_i F(x) = 0 \quad (1 \leq i \leq n).$$

This is a system of the preceding kind, so there were at most $(2q+1)^n$ chosen zeros. The same finite-subset argument proves the claim. □

Proof of Lemma 5.3. We use generic level perturbations and Lagrange multipliers to find m-dimensional pieces whose images cover almost all of the image. Their coordinate fibres are controlled by Lemma A.1.[1]

If Y is empty the assertion is immediate. If $k < m$, the image has m-volume zero because a polynomial map is locally Lipschitz. Assume henceforth that $1 \leq m \leq k$, and write

$$B = \sup_{x\in Y} J_m\phi(x), \qquad q = \max\{2, \Delta_0\}.$$

[1] An earlier version of these companion notes used the quantitative approximate definable choice theorem of Lerario, Rizzi and Tiberio [LRT25, Theorem 44]. The direct argument here suffices for the polynomial maps used in these notes.

After translating the source ball, suppose it is centred at zero. Replace each equality in the description of Y by two opposite weak inequalities, and add the redundant inequality $R_0^2 - \|x\|^2 \geq 0$. We obtain polynomials $g_1, \ldots, g_T$, of degrees at most q, such that

$$Y = \{x : g_t(x) \geq 0 \text{ for all } t\}, \qquad T \leq 2\Delta_0 + 1.$$

Outer perturbation. Fix $\eta > 0$. By continuity of the Jacobian, $J_m\phi < B + \eta$ on an open neighbourhood U of Y. For $b = (b_1, \ldots, b_T)$ put

$$Y_b = \{x : g_t(x) \geq b_t \text{ for all } t\}.$$

For some $\delta > 0$, every $b \in (-\delta, 0)^T$ satisfies

$$Y \subseteq Y_b \subseteq U \cap \overline{B}(0, R_0 + 1), \qquad Y_b \text{ compact.} \tag{47}$$

Indeed, the ball inequality bounds all these sets when δ is small. If containment in U failed for arbitrarily small δ, a convergent subsequence of counterexamples would have a limit in $\{g_t \geq 0 \text{ for all } t\} = Y$ outside U.

Generic parameters. For $J \subseteq \{1, \ldots, T\}$, write $j = |J|$ and $g_J = (g_t)_{t \in J}$. Introduce multiplier variables $\lambda \in \mathbb{R}^m$ and $\mu \in \mathbb{R}^j$, and define polynomial maps

$$\begin{aligned} G_J(x) &= (g_J(x), \phi(x)), \\ H_J(x, \lambda, \mu) &= (g_J(x),\ D\phi(x)^*\lambda + Dg_J(x)^*\mu)\,, \\ K_{J,I}(x, \lambda, \mu) &= (H_J(x, \lambda, \mu), \pi_I x)\,, \end{aligned}$$

where I ranges over the m-element subsets of $\{1, \ldots, k\}$ and π_I is coordinate projection onto $\mathbb{R}^m$. The source of H_J has dimension $k + m + j$ and its target dimension is $k + j$; $K_{J,I}$ has equal source and target dimensions $k + m + j$. Every coordinate of these maps has degree at most q: differentiating lowers degree by one, and multiplication by one multiplier restores at most that degree.

Sard's theorem and Fubini's theorem allow us to choose $b \in (-\delta, 0)^T$ and $v \in \mathbb{R}^k$ with the following properties, simultaneously for every J and I:

1. (b_J, y) is a regular value of G_J for almost every $y \in \mathbb{R}^m$;
2. (b_J, v) is a regular value of H_J;
3. $((b_J, v), z)$ is a regular value of $K_{J,I}$ for almost every $z \in \mathbb{R}^m$.

For clarity, apply Sard to each displayed map, then slice its null set of critical values in the indicated parameter order. A full-measure condition on b_J lifts to one on the full vector b; the same holds for (b_J, v). There are only finitely many J and I, so their full-measure sets have a common intersection, which meets $(-\delta, 0)^T \times \mathbb{R}^k$.

Lagrange pieces cover the image. Let $A_J(x)$ be the $k \times (m + j)$ matrix with columns $\nabla\phi_1(x), \ldots, \nabla\phi_m(x)$ and $(\nabla g_t(x))_{t \in J}$. Restrict the regular level set $H_J^{-1}(b_J, v)$ to the open locus where these columns are linearly independent, and denote the resulting smooth m-manifold by W_J. Its projection onto the x coordinates is injective: the multipliers are uniquely determined by $A_J(x)(\lambda, \mu) = v$. The projection is also an immersion. A tangent vector in its kernel has $\dot{x} = 0$, and differentiating $H_J = (b_J, v)$ then gives $A_J(x)(\dot{\lambda}, \dot{\mu}) = 0$, hence $\dot{\lambda} = \dot{\mu} = 0$. Moreover, the inverse on its image is smooth, since the multipliers are given by

$$(\lambda, \mu) = \big(A_J(x)^* A_J(x)\big)^{-1} A_J(x)^* v.$$

Thus the projected set M_J is an embedded smooth m-manifold. It is empty if $m + j > k$, since $A_J(x)$ cannot then have independent columns. Set $C_J = M_J \cap Y_b$. It is a Borel subset of M_J,

and is covered by countably many smooth coordinate charts; the area formula therefore applies to C_J.

Exclude the null union of exceptional output values in property (1). For a remaining $y \in \phi(Y_b)$, maximize $v \cdot x$ on the nonempty compact fibre $Y_b \cap \phi^{-1}(y)$, and let x be a maximizer. Take J to be exactly the active constraints, those satisfying $g_t(x) = b_t$. Regularity of (b_J, y) for G_J says that $A_J(x)$ has independent columns. Near x, the equality manifold $g_J = b_J$, $\phi = y$ lies in Y_b, because the other inequalities are strict at x. The Lagrange multiplier rule on this manifold gives $v = D\phi(x)^*\lambda + Dg_J(x)^*\mu$. Consequently $x \in C_J$, and

$$\phi(Y_b) \subseteq \bigcup_J \phi(C_J) \quad \text{up to a set of } m\text{-volume zero.} \tag{48}$$

Coordinate fibres have bounded multiplicity. Fix J, I, and a value z as in property (3). The fibre $K_{J,I}^{-1}((b_J, v), z)$ is the zero set of a nonnegative polynomial: sum the squares of its $k + m + j$ defining equations. Its degree is at most $2q$. Since this is a regular fibre of a map between spaces of equal dimension, every point is isolated, by the inverse function theorem. Lemma A.1 gives

$$\#K_{J,I}^{-1}((b_J, v), z) \le (2q+1)^{k+m+j} =: L_j.$$

Every point of $C_J \cap \pi_I^{-1}(z)$ has a multiplier lift into this fibre, so

$$\#(C_J \cap \pi_I^{-1}(z)) \le L_j \quad \text{for almost every } z.$$

No boundedness of the multiplier variables is required.

Area and Jacobian bounds. At a point of M_J, let j_I be the m-Jacobian of $\pi_I|_{M_J}$. Cauchy–Binet gives $\sum_I j_I^2 = 1$, and hence $\sum_I j_I \ge 1$. Using the area formula [Fed69, 3.2.20] for each coordinate projection, and the radius bound in (47), we obtain

$$\begin{aligned}
\mathcal{H}^m(C_J) &\le \sum_I \int_{C_J} j_I \, d\mathcal{H}^m \\
&= \sum_I \int_{\mathbb{R}^m} \#(C_J \cap \pi_I^{-1}(z)) \, dz \\
&\le \binom{k}{m} L_j \, \omega_m (R_0 + 1)^m.
\end{aligned}$$

Apply the area formula once more, now to $\phi|_{M_J}$. Its restricted Jacobian is at most the ambient m-Jacobian, so

$$\mathrm{Vol}_m(\phi(C_J)) \le (B + \eta)\mathcal{H}^m(C_J).$$

By $Y \subseteq Y_b$ and (48), sum over the active constraint sets J. Only $j = |J| \le k - m$ can contribute, and there are $\binom{T}{j}$ subsets of size j. It follows that

$$\mathrm{Vol}_m(\phi(Y)) \le \binom{k}{m} (2q+1)^{k+m} (R_0+1)^m \, \omega_m (B + \eta) \sum_{j=0}^{\min\{T, k-m\}} \binom{T}{j} (2q+1)^j.$$

The parameters b, v may depend on η, but this coefficient does not. Letting $\eta \downarrow 0$ proves the explicit bound

$$\mathrm{Vol}_m(\phi(Y)) \le \binom{k}{m} (2q+1)^{k+m} (R_0+1)^m \, \omega_m \sup_Y J_m\phi \sum_{j=0}^{\min\{T, k-m\}} \binom{T}{j} (2q+1)^j. \tag{49}$$

Finally, $\binom{k}{m} \le 2^k$ and the sum is at most $(2q+2)^T$, with T, q bounded in terms of Δ_0 alone. Absorbing these fixed factors and the radius factor into C_Δ^{k+m} proves Lemma 5.3. □

A.2 The volume of the unitary group at its natural scale

The unitary group $\mathrm{U}(d^2)$ has Frobenius diameter of order d. The following lemma compares the volume of a neighbourhood of a subset with its Haar measure at this scale.

Lemma A.2 (Normal volume)**.** *Let $d \geq 2$ and $N = d^4$. There are universal $c, C > 0$ such that $\mathrm{Vol}_F\,\mathrm{U}(d^2) \geq (cd)^N \omega_N$, and for every measurable $\mathcal{S} \subseteq \mathrm{U}(d^2)$ and $0 < t \leq d/64$,*

$$\mathrm{Vol}_{2N}\big(U_t(\mathcal{S})\big) \geq c^N \omega_N\, t^N\ \mathrm{Vol}_F\,\mathrm{U}(d^2)\,\mu(\mathcal{S}). \tag{50}$$

Proof. First, at least a quarter of the Hermitian directions Q of Frobenius norm one satisfy $\|Q\|_{\mathrm{op}} \leq 32/d$. To see this, let G be a standard Gaussian vector in the N-dimensional real space of Hermitian $d^2 \times d^2$ matrices. For every complex unit vector v, $v^\dagger G v$ is a standard real Gaussian. A $\frac{1}{4}$-net of the complex unit sphere of $\mathbb{C}^{d^2}$ has at most 9^{2d^2} points, and $\|G\|_{\mathrm{op}} \leq 2\max_v |v^\dagger G v|$ over the net. Hence $\Pr(\|G\|_{\mathrm{op}} > 16d) \leq 2 \cdot 9^{2d^2} e^{-32d^2} < \frac{1}{4}$. For $X = \|G\|_F^2$ we have $\mathbb{E}X = N$ and $\mathbb{E}X^2 = N(N+2)$, and the Paley–Zygmund inequality gives $\Pr(X \geq N/4) \geq \frac{9}{16}\frac{N}{N+2} \geq \frac{1}{2}$ since $N \geq 16$. By the union bound, both events occur with probability at least $1 - \frac{1}{4} - \frac{1}{2} = \frac{1}{4}$, and then $\|G\|_{\mathrm{op}}/\|G\|_F \leq 32/d$. Since the direction of G is uniform, this proves the claim, and consequently

$$\mathrm{Vol}\{Q = Q^\dagger : \|Q\|_F < t,\ \|Q\|_{\mathrm{op}} \leq \tfrac{1}{2}\} \geq \tfrac{1}{4}\,\omega_N t^N \qquad (t \leq d/64).$$

For such Q, the map $(U, Q) \mapsto U(I+Q)$ is injective by uniqueness of the polar decomposition, and $\|U(I+Q) - U\|_F = \|Q\|_F$. After left translation its differential is $(A, Q') \mapsto A + Q' + AQ$, with A anti-Hermitian and Q' Hermitian. The map $(A, Q') \mapsto A + Q'$ is an isometry, and the perturbation has norm at most $\|A\|_F/2$, so the Jacobian is at least 4^{-N}. Integrating over $\mathcal{S}$ and the set of Q above proves (50).

For the volume of the group, use the Cayley chart $A \mapsto (I+A)(I-A)^{-1}$ on anti-Hermitian A with $\|A\|_{\mathrm{op}} \leq 1$. It is injective, and its derivative $B \mapsto 2(I-A)^{-1}B(I-A)^{-1}$ has all singular values at least $\frac{1}{2}$. By the same angular estimate, at least a quarter of the Frobenius ball of radius $d/32$ in the anti-Hermitian matrices lies in the domain, so the image has volume at least $2^{-N} \cdot \frac{1}{4}\,\omega_N (d/32)^N$. □

A.3 Thickening the image

Proof of Proposition 5.4. Use ordinary Frobenius norms and put

$$\Theta_s(x, Y') = \widehat{T}(x) + 2d\,s\,Y', \qquad x \in \widetilde{\mathcal{Z}},\ Y' \in \mathbb{C}^{d^2 \times d^2},\ \|Y'\|_F \leq 1.$$

The normalized bounds L and λ become the ordinary bounds dL and $d\lambda$. Since $\mathrm{d}\Theta_s\,\mathrm{d}\Theta_s^* = \mathrm{d}\widehat{T}\,\mathrm{d}\widehat{T}^* + 4d^2 s^2 I$, the singular values of $\mathrm{d}\Theta_s$ are at most $\sigma_j(\mathrm{d}\widehat{T}) + 2ds$, so

$$J_{2N}\Theta_s \leq d^{2N}(L + 2s)^\ell (\lambda + 2s)^{2N-\ell}. \tag{51}$$

The source has universal radius and is defined by the constraints for $\widetilde{\mathcal{Z}}$ and one additional quadratic inequality. The map Θ_s is polynomial of the same bounded degree as $\widehat{T}$, so their polynomial format is bounded in terms of Δ_0, with one additional constraint, in $D + 2N$ source variables. This fixed enlargement of the format budget is absorbed into C_Δ. The image of Θ_s is the neighbourhood $U_{2ds}(\widehat{T}(\widetilde{\mathcal{Z}}))$, so Lemma 5.3 gives

$$\mathrm{Vol}_{2N}\big(U_{2ds}(\widehat{T}(\widetilde{\mathcal{Z}}))\big) \leq e^{C_\Delta(D+N)}\,\omega_{2N}\,d^{2N}(L + 2s)^\ell(\lambda + 2s)^{2N-\ell}.$$

Every point of $\mathcal{S}$ is within Frobenius distance $d\tau \leq ds$ of $\widehat{T}(\widetilde{\mathcal{Z}})$, so $U_{ds}(\mathcal{S}) \subseteq U_{2ds}(\widehat{T}(\widetilde{\mathcal{Z}}))$. Lemma A.2 with $t = ds \leq d/64$, that is $s \leq c_0 := 1/64$, gives

$$\mathrm{Vol}_{2N}\big(U_{ds}(\mathcal{S})\big) \geq c^N \omega_N^2\, d^{2N} s^N \mu(\mathcal{S}).$$

Now $\omega_{2N} \le \omega_N^2$, $\lambda \le s$, and the factors d^{2N} cancel. The remaining power of s is $s^{2N-\ell}/s^N = s^{N-\ell}$, which proves the proposition. A union of e^{CD} sets changes the constant in the exponential only. □

B Schmidt-rank floors

These elementary bounds verify the size conditions on K used in Corollaries 5.6, 6.4 and 6.8, and they record constant-error bounds that do not depend on the tube estimate. For exact unitaries, the first is the familiar requirement that the available Schmidt rank must cover the operator-Schmidt rank of the gate; Stahlke and Griffiths prove the resource-only version for separable operations [SG11]. In NLQC, quantum messages also carry Schmidt rank across the cut, which is why the bound is in terms of $K = r\, m_A m_B$.

Lemma B.1 (Schmidt-rank floors)**.** *Let a protocol of footprint at most K have score deficit at most $\varepsilon < 1$.*

(i) For a unitary target U, $\Lambda_K(|U\rangle\!\rangle) \ge 1 - \varepsilon$, where $|U\rangle\!\rangle$ is split by the laboratory cut.

(ii) For a PVM with basis $(\varphi_i)_{i \le d^2}$, there are integers $k_i \ge 0$ with

$$\sum_i k_i \le K, \qquad \sum_i \Lambda_{k_i}(\varphi_i) \ge d^2(1-\varepsilon)^2.$$

Consequently, approximating SWAP *requires $K \ge d^2(1-\varepsilon)$, every basis requires $K \ge d^2(1-\varepsilon)^2$, and a complete basis of maximally entangled vectors requires $K \ge d^3(1-\varepsilon)^2$. At $\varepsilon = 0$, (i) gives that K is at least the operator-Schmidt rank of U, and (ii) gives $K \ge \sum_i \mathrm{SR}(\varphi_i)$. Moreover:*

(iii) If U has full operator-Schmidt rank d^2, which holds for Haar-almost every U, then $K \ge d^2$ whenever $\varepsilon < \varpi_{\min}(U)$, the smallest squared Schmidt coefficient of $|U\rangle\!\rangle$. If every basis vector φ_i has Schmidt rank d, which holds for almost every PVM, then $K \ge d^3$ whenever $\varepsilon < \lambda_*/(2d^2)$, where λ_* is the smallest squared Schmidt coefficient of all the φ_i.

Proof. Select the pure resource component and use the notation of the proof of Lemma 3.2. (i) There, $1 - \varepsilon \le p \le \Lambda_K(|U\rangle\!\rangle \otimes \gamma') \le \Lambda_K(|U\rangle\!\rangle)$ by (12). (ii) There, $\Lambda_K(v_0) \ge p^2 \ge (1-\varepsilon)^2$. Let k_i be the number of selected triples (i,j,k) in block i of (17), and let $\mathcal{J}_i$ be the set of indices j that occur in them, so $|\mathcal{J}_i| \le k_i$. Since $\sum_k \zeta_{ik}^2 = \|\tilde{\gamma}_i\|^2 \le 1$, the selected mass in block i is at most $d^{-2}\sum_{j\in\mathcal{J}_i} \lambda_{ij} \le d^{-2}\Lambda_{k_i}(\varphi_i)$. Summing over i proves (ii).

The normalized operator-Schmidt spectrum of SWAP consists of d^2 equal weights, so $\Lambda_K(|\mathsf{SWAP}\rangle\!\rangle) = \min\{K/d^2, 1\}$. For every unit vector, $\Lambda_0 = 0$ and $\Lambda_k \le 1$, so $\sum_i \Lambda_{k_i}(\varphi_i) \le |\{i : k_i \ge 1\}| \le \sum_i k_i \le K$. A maximally entangled vector has $\Lambda_k \le k/d$, so $\sum_i \Lambda_{k_i}(\varphi_i) \le K/d$.

For (iii), if $K < d^2$ then $\Lambda_K(|U\rangle\!\rangle) \le 1 - \varpi_{\min}(U)$, so (i) forces $\varepsilon \ge \varpi_{\min}(U)$. If $\sum_i k_i < d^3$, then $k_i < d$ for some i, so $\sum_i \Lambda_{k_i}(\varphi_i) \le d^2 - \lambda_*$, and (ii) forces $(1-\varepsilon)^2 \le 1 - \lambda_*/d^2$, which fails for $\varepsilon < \lambda_*/(2d^2)$. The determinant of the realignment of U, and the product over i of the determinants of the $d \times d$ coefficient matrices of the columns of a basis matrix, are real-analytic functions on $\mathrm{U}(d^2)$ that do not vanish identically (for instance at SWAP and at a basis of maximally entangled vectors), so their zero sets are null. □

For $0 < \varepsilon \le 1/2$, universal approximation of unitaries thus forces $K \ge d^2/2$, and universal approximation of PVMs forces $K \ge d^2/4$ already because of a single basis, and in fact $K \ge d^3/4$. The precision term $c\, d\sqrt{\ln(1/\varepsilon)}$ of Theorem A exceeds these floors only when $\ln(1/\varepsilon)$ is of order d^2 for unitaries and d^4 for measurements, while the term $c\, d^2\sqrt{\ln(1/\varepsilon)}$ of Theorem B adds to the floor d^2 for unitaries at every error, and exceeds the floor d^3 for measurements once $\ln(1/\varepsilon)$ is of order d^2.

C An effective bound for an explicit controlled-phase gate

The main paper [FGMS26] proves that $C_1 = \mathrm{diag}(1,1,1,e^i)$ has no exact protocol with finite-dimensional resources: its operator purity $P(C_1) = (3+\cos 1)/4$ is transcendental, whereas the purity of every exactly implementable unitary is algebraic. By Proposition 2.2, the least error $\varepsilon_K(C_1)$ at footprint K is positive, but that compactness argument gives no rate. The almost-everywhere results also give no bound for this named target. Here we make the arithmetic obstruction effective, using quantifier elimination and a transcendence measure. Keeping classical outcomes explicit gives a smaller optimization problem and a stronger quantum-footprint bound. Throughout this appendix, $d = 2$.

Theorem C.1 (Effective bound for C_1)**.** *There are constants $C_E, C_{E,\mathrm{q}} > 0$ with the following property. For $K \geq 1$ and $0 < \varepsilon < 1$, every protocol of footprint at most K that implements C_1 with $\delta_{\mathrm{Choi}} \leq \varepsilon$ satisfies*

$$\varepsilon \geq \exp\bigl[-\exp(C_E K^2)\bigr]. \tag{52}$$

If classical messages are free as in Section 7, every implementation of C_1 with $\delta_{\mathrm{Choi}} \leq \varepsilon$, resource Schmidt number at most R and quantum footprint at most K_{q} satisfies

$$\varepsilon \geq \exp\bigl[-\exp(C_{E,\mathrm{q}} R^4 K_{\mathrm{q}}^2)\bigr]. \tag{53}$$

In particular, every protocol of quantum footprint at most K_{q} obeys $\varepsilon \geq \exp[-\exp(C_{E,\mathrm{q}} K_{\mathrm{q}}^6)]$. Consequently, for $0 < \varepsilon < 1/e$,

$$\log_2 K_\varepsilon(C_1) \geq \tfrac{1}{2}\log_2 \ln\ln(1/\varepsilon) - O(1), \qquad \log_2 K_{\mathrm{q},\varepsilon}(C_1) \geq \tfrac{1}{6}\log_2 \ln\ln(1/\varepsilon) - O(1),$$

where $K_{\mathrm{q},\varepsilon}$ is the least quantum footprint. In LOSCC this gives at least $\frac{1}{3}\log_2 \ln\ln(1/\varepsilon) - O(1)$ total initial resource qubits.

Proof of Theorem C.1. For $-1 \leq c \leq 1$, $s \geq 0$ and $c^2 + s^2 = 1$, let $C(c,s) = \mathrm{diag}(1,1,1,c+is)$, so that $C_1 = C(c_*, s_*)$ with $c_* = \cos 1$ and $s_* = \sin 1$.

Step 1: the least deficit as a compact polynomial optimization. Fix a factorization $r\, m_A m_B \leq K$. The score is affine in the resource, so it suffices to consider pure resources on $\mathbb{C}^r \otimes \mathbb{C}^r$. By Lemma 4.1, the kept register of party X has dimension at most $2rm_X$, and the input of its final channel has dimension at most $2K$. The encoders are isometries. At the end we discard everything except the logical output, and represent each final channel, from a space of dimension at most $2K$ to the fixed output $\mathbb{C}^2$, by a Stinespring isometry with environment of dimension at most $4K$. The resource, the two encoders and the two final isometries then use

$$N_{\mathrm{var}} \leq CK^2 \tag{54}$$

real coordinates. For each choice of the finitely many discrete dimensions, the normalization and isometry equations define a compact real algebraic set $\mathcal{Z}$ over $\mathbb{Q}$, the analogue of the strategy sets $\mathcal{Z}_\nu$ of [FGMS26] without the exactness constraint. In real coordinates, the score $q_{C(c,s)}$ is a polynomial with rational coefficients of fixed degree in (c,s,x). We package all constraints for one dimension choice into one sum of squares; with the fixed list of score and box inequalities, the number and degrees of the polynomials are fixed, and after clearing denominators their integer coefficients have bit size polynomial in K. Let $\mathcal{F}_K$ be the resulting finite collection, and put $f = 1 - q_{C(c,s)}$ and

$$g_K(c) = \min_{\mathcal{Z} \in \mathcal{F}_K} \ \min\bigl\{ f(c,s,x) : x \in \mathcal{Z},\ s \geq 0,\ c^2 + s^2 = 1 \bigr\}. \tag{55}$$

All minima are attained, and $0 \leq g_K \leq 1$. All reductions above are explicit, so the constants in these format bounds can be computed.

A protocol with score one implements its target exactly (Proposition 2.2), and C_1 has no exact protocol with finite-dimensional resources [FGMS26]. Hence

$$g_* := g_K(c_*) > 0,$$

and a protocol of footprint at most K with $\delta_{\mathrm{Choi}} \le \varepsilon$ for C_1 gives $g_* \le \varepsilon$.

Step 2: elimination and transcendence estimates. One-block quantifier elimination over the reals turns an existential formula with N_{var} quantified variables and two free variables, built from a fixed number of integer polynomials of degree at most d_0 and coefficient bit size at most τ, into a quantifier-free formula whose polynomials Q satisfy

$$\deg Q \le d_0^{O(N_{\mathrm{var}})}, \qquad \mathrm{bitsize}(Q) \le \tau\, d_0^{O(N_{\mathrm{var}})} \tag{56}$$

[BPR06, Theorem 14.16]. In general these exponents also grow with the number of free variables and of quantifier blocks; here there is a single block and the number of free variables is fixed at two, so only the dependence on N_{var} remains. Cijsouw's transcendence measure for e^β [Cij74, Theorem 1] implies that for every nonzero algebraic β there is an effectively computable $C_\beta > 0$ such that every nonconstant integer polynomial S of degree ℓ and coefficient height H satisfies

$$|S(e^\beta)| \ge \exp\bigl[-C_\beta\, \ell^2(\ell + \ln H)\bigr]. \tag{57}$$

Step 3: an algebraic boundary near the target. Choose $\mathcal{Z}_* \in \mathcal{F}_K$ that attains g_*, and consider the epigraph

$$E = \bigl\{(c, y) \in [-1, 1] \times [0, 1] : \exists\, s \ge 0,\ x \in \mathcal{Z}_* \text{ with } c^2 + s^2 = 1,\ f(c, s, x) \le y\bigr\}.$$

The sine coordinate is quantified out, and only c is kept. By (54) and (56), E is a Boolean combination of sign conditions on nonzero integer polynomials $Q(c, y)$ of degree L and bit size τ_Q with

$$L,\ \tau_Q \le \exp(A_0 K^2) \tag{58}$$

for an effective constant A_0. The point (c_*, g_*) lies on the boundary of E: it belongs to E, while $(c_*, g_* - t) \notin E$ for $0 < t < g_*$. Hence some polynomial Q of the formula vanishes at (c_*, g_*); otherwise all signs, and hence membership in E, would be locally constant. Divide Q by the largest power of y that divides it; since $g_* > 0$, the quotient still vanishes at (c_*, g_*), and we rename it Q. Then $Q_0(c) := Q(c, 0)$ is a nonzero integer polynomial.

Step 4: transcendence separates the target from the boundary. Let Q_0 have degree $\ell \le L$ and height $H \le H_Q \le 2^{\tau_Q}$. If $\ell \ge 1$, then $S(z) = 2^\ell z^\ell Q_0\bigl(\frac{z+z^{-1}}{2}\bigr)$ is a nonconstant integer polynomial of degree at most 2ℓ and height at most $(\ell + 1)2^\ell H$, and $|Q_0(\cos 1)| = 2^{-\ell}|S(e^i)|$. Applying (57) with $\beta = i$, and noting that a nonzero constant Q_0 has absolute value at least one,

$$|Q_0(c_*)| \ge \exp\bigl[-C'(1 + L)^2(1 + L + \tau_Q)\bigr].$$

On the other hand, the mean value theorem in y gives

$$|Q_0(c_*)| = |Q(c_*, 0) - Q(c_*, g_*)| \le g_*(L + 1)^3 H_Q,$$

where the last factor bounds $|\partial_y Q|$ on $[-1, 1] \times [0, 1]$. Absorbing this factor into the exponential gives, after increasing C',

$$g_* \ge \exp\bigl[-C'(1 + L)^2(1 + L + \tau_Q)\bigr]. \tag{59}$$

With (58), this gives an effective $C_E > 0$ with $g_* \ge \exp[-\exp(C_E K^2)]$, and hence (52).

Step 5: free classical messages. By Lemma 7.1, select a pure resource of Schmidt rank $r \le R$ and finite classical outcomes without decreasing the score. Write $a = m_A^{\mathrm{q}}$, $b = m_B^{\mathrm{q}}$ and $q = rab \le K_{\mathrm{q}}$, and pad the outcome sets to $L_A = L_B = 4r^2$.

Purify each encoder branch, keeping its Kraus environment locally. Expanding its images over an input basis and a message basis restricts its kept space to dimension at most $2ra$ for Alice and $2rb$ for Bob. The embedding of each kept support into the original kept space and purification environment can be absorbed into the conditional decoder channel, which discards that environment; extend this channel arbitrarily on unused padding. Thus the branch maps have the form

$$V_{A,x} : \mathbb{C}^{2r} \longrightarrow \mathbb{C}^{2ra} \otimes \mathbb{C}^{a}, \qquad V_{B,y} : \mathbb{C}^{2r} \longrightarrow \mathbb{C}^{2rb} \otimes \mathbb{C}^{b},$$

with $\sum_x V_{A,x}^\dagger V_{A,x} = \sum_y V_{B,y}^\dagger V_{B,y} = I_{2r}$. After exchange, each decoder input has dimension $2q$. For every transcript (x, y), its two conditional channels admit isometries

$$W_{A,xy}, W_{B,xy} : \mathbb{C}^{2q} \longrightarrow \mathbb{C}^{2} \otimes \mathbb{C}^{4q}.$$

Together with the resource vector in $\mathbb{C}^r \otimes \mathbb{C}^r$, these maps use

$$N_{\mathrm{q}} = 2r^2 + 32r^4(a^2 + b^2) + 1024r^4q^2 \le 1090r^4q^2 \tag{60}$$

real coordinates. The resource normalization, stacked instrument normalizations and decoder isometry equations bound every entry and define a compact real algebraic set over $\mathbb{Q}$. Conversely, every feasible tuple is a physical finite protocol; zero branches are allowed. The quadratic equations can again be combined into one degree-four sum of squares.

Let A_{xyuv} be the Kraus operator obtained by projecting the two decoder environments onto basis vectors u, v. The score is

$$q_{C(c,s)} = \frac{1}{16} \sum_{x,y,u,v} |\mathrm{Tr}(C(c, s)^\dagger A_{xyuv})|^2.$$

Each Kraus amplitude contains one resource entry, two encoder entries and two decoder entries, so this score has degree at most twelve including (c, s). All index ranges are polynomially bounded in q, hence clearing the fixed denominator gives coefficient bit size polynomial in q. Classical transcripts contribute separate summands.

For each fixed (r, a, b), minimize the deficit over this compact family. At c_* the minimum is positive by exact impossibility, and at most ε by score selection. Applying (56) to its epigraph gives $L, \tau_Q \le \exp(Cr^4q^2)$; then (59) yields

$$\varepsilon \ge \exp[-\exp(Cr^4q^2)] \ge \exp[-\exp(C_{E,\mathrm{q}}R^4K_{\mathrm{q}}^2)].$$

The constants are uniform in (r, a, b), so no count of architectures is needed. Since $r \le q \le K_{\mathrm{q}}$, the sixth-power bound follows. Shared randomness under uniform budgets is handled by first selecting a seed whose score is at least the average. Taking logarithms gives the stated footprint bounds; in LOSCC, a resource of Schmidt number R occupies at least $2\log_2 R$ qubits. □

Remark C.2 (Direct elimination). The degree and height bounds needed here can also be obtained without general quantifier elimination. Write $P(x) = 0$ for the sum-of-squares constraint and $S(c, s, x) = 16q_{C(c,s)}$ for the integer score polynomial, with k real protocol coordinates. Both have degree at most twelve, and an optimal feasible point has coordinates in $[-1, 1]$. If their coefficient masses (sums of absolute coefficients) are at most H, choose an integer $\tau \ge 1$ with $32 + H < 2^\tau$. There is a nonzero $A \in \mathbb{Z}[g, c]$ vanishing at $(g_*, \cos\theta)$ with

$$\deg A \le 12^{4(k+2)}, \qquad \max_\alpha |A_\alpha| < 2^{\tau 12^{4(k+2)}}.$$

For the construction, deform $P^2 + (y - S)^2$ to $\sum_i x_i^{26} + y^{26} - (k + 16^{26} + 1) + \lambda(P^2 + (y - S)^2)$. Maximizing y on its compact zero set gives a critical system with leading powers x_i^{25} and y^{26}. Reduction to $26 \cdot 25^k$ monomials gives a nonzero determinant eliminant. Taking its norm under

$s^2 = 1 - c^2$, then its leading coefficient in λ and the limit $\lambda \to \infty$, yields the relation above after $y = 16(1 - g)$. This supplies precisely the elimination estimate required for the final double-exponential rate, for both the charged and transcript parametrizations. It is the route used in the formal proof.

Real quantifier elimination has been used before to show decidability of compact, fixed-dimensional questions in quantum information [WCPG11]; here the explicit degree and height bounds are what make the gap effective.

The bound is weaker than the generic result of Theorem 8.1: for almost every controlled phase, $\log_2 K_\varepsilon$ grows at least as $\frac{1}{2}\log_2 \ln(1/\varepsilon) - O(1)$, whereas the explicit bound has the additional logarithm. In LOSCC this gives a triple-logarithmic lower bound on the total resource-qubit cost, since a resource of Schmidt number r occupies at least $2\log_2 r$ qubits. It does not give a lower bound on entanglement entropy.

The same proof applies to C_θ for every nonzero real algebraic θ, with constants depending on θ: the exact obstruction uses Hermite–Lindemann, and Cijsouw's measure applies with $\beta = i\theta$. If $\sin\theta < 0$, complex conjugating the target and all protocol maps preserves the score and the footprint and reduces to the upper semicircle used in the proof.

D Explicit constants in the universal bounds

We record the numerical estimates behind the universal bounds and their scalar consequences. This is a compressed derivation; the full checked calculations are in the Lean development described in Section 9. Throughout, K is the charged footprint, $d \geq 2$, $0 < \varepsilon \leq 1/2$, $L = \ln(1/\varepsilon)$, and $N = d^4$. Universality allows a different protocol for each target. The bounds hold for pure and finite mixed resources, with the same local maps on every component. Unitary error is Choi infidelity or normalized diamond error; measurement error is the score deficit or worst-case joint total variation for ordered rank-one PVMs.

Numerical bounds. Writing $x_+ = \max\{x, 0\}$, the unitary bounds are

$$K \geq d^2 \max\left\{1 - \varepsilon, \frac{\sqrt{L}}{51}, \frac{\sqrt{(L - 420)_+}}{46}\right\}, \qquad d \geq 2, \tag{61}$$

$$K \geq d^2 \max\left\{1 - \varepsilon, \frac{\sqrt{L}}{24}, \frac{\sqrt{(L - 92)_+}}{22}\right\}, \qquad d \geq 8. \tag{62}$$

For measurements,

$$K \geq \max\left\{d^3(1 - \varepsilon)^2, \frac{d^2\sqrt{L}}{a_d}\right\}, \qquad a_d = \begin{cases} 36, & d = 2, \\ 27, & d \geq 3. \end{cases} \tag{63}$$

At $d = 2^n$, the clean precision term in either bound gives $\log_2 K \geq 2n + \frac{1}{2}\log_2 L - 6$. The offset improves to 5 for unitaries when $n \geq 3$ and for measurements when $n \geq 2$; measurements also satisfy $\log_2 K \geq 3n - 2$. These constants leave the resource exponents unchanged. The spectral floors dominate until the requested error is very small.

Volume estimates. The numerical calculation uses an alternative barrier proof of the image-volume estimate. Thicken the polynomial equalities into one strict sum-of-squares inequality, shift the remaining inequalities, and multiply these factors with a ball constraint. On each nonempty fibre meeting the open neighbourhood, a generic function $\phi(x)e^{\langle \ell, x\rangle}$, where ϕ is this positive barrier, has an interior maximum. Its Lagrange equations provide image-covering charts; polynomial zero counts bound their multiplicities. For the witness families here, equality

degrees are at most 12, the sum of inequality degrees is at most 14, and the map degree is at most 18. The barrier degree is therefore at most 40. The resulting volume factor, multiplying the unit-ball volume ω_{2N} and Jacobian bound, is

$$2^{P+2N}82^{P+4N}10^{N}, \qquad P \le 38K^2 \text{ (unitaries)}, \quad P \le 42K^2 \text{ (measurements)}.$$

Thus the factor per source coordinate is 164. Retaining its logarithm, and keeping source and target dimensions separate, avoids the large losses of a single unspecified exponential constant. The near-Bell coordinate count uses $d^3 \le 2K$, as ensured in the precision-dominant regime below.

Unitary reduction. The refined restricted Haar estimate on the near-SWAP patch holds with either applicable choice below:

$$\begin{aligned}\mu(\mathcal{B}_d \cap \mathsf{Reach}^{\mathrm{u}}_{K,\varepsilon}) &\le \min\{1, \exp(AK^2 + BN)\varepsilon^{\kappa N}\},\\ (A, B, \kappa) &= (595/3, 687/20, 3/32), \qquad d \ge 2,\\ (A, B, \kappa) &= (595/3, 703/20, 7/16), \qquad d \ge 8.\end{aligned}$$

Here $K \ge d^2/2$; Haar measure is not renormalized on the patch (and diamond reachability uses outer measure). In particular, the constant $C = 336$ is valid in Theorem 6.3. The patch mass is at least e^{-5N}, so universality gives $\kappa NL \le AK^2 + (B+5)N$. Together with $K \ge d^2(1-\varepsilon)$, this yields $K \ge cd^2\sqrt{L}$ whenever

$$c^2 \le 1/24, \qquad c^2\big(A + \tfrac{101}{100}(B+5)\big) \le \kappa.$$

Indeed, if the precision term exceeds the floor, then $\varepsilon < 1/403$ and $N \le (101/100)K^2$. Choose $c = 1/51$ or $1/24$. For the shifted bounds, use $c'^2 A \le \kappa$ and $B + 5 \le \kappa t$ with $(c', t) = (1/46, 420)$ or $(1/22, 92)$.

Measurement reduction. Put $t = 3d^2 - 2$ and $P = 42K^2$. For $c = 1/36$, or $c = 1/27$ when $d \ge 3$, if $cd^2\sqrt{L}$ exceeds the cubic floor, then $\varepsilon < 1/403$ and $K \ge 19d^3/20$. For tube radius $\sqrt{50\varepsilon/9}\,P \le 1/2$, comparison with the near-Bell patch mass $(1 + 8d^{3/2})^{-2N}$ gives

$$\frac{N-t}{2}L \le 239K^2, \qquad \frac{N-t}{2} \ge \begin{cases} 3N/16, & d \ge 2,\\ N/3, & d \ge 3.\end{cases}$$

For larger radius, $NL \le 11K^2$ instead. Both cases give the stated precision term: take $c = 1/36$ or $1/27$, respectively, and check $c^2 \le 1/24$, $11c^2 \le 1$, and $239c^2 \le 3/16$ or $1/3$.